

Relative Skills in the Classroom: Teachers' Gender-Differentiated Impacts on Test Scores and Course Grades

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Abstract

Gender gaps in academic achievement vary across subjects and assessment types: girls outperform boys on teacher-assigned course grades and on standardized reading tests, while gaps in math test scores favor boys in early grades but reverse by middle school. I investigate how teachers' impacts on these outcomes differ for boys and girls, and what explains the heterogeneity. Using longitudinal administrative data from North Carolina, I estimate value-added measures for fifth-grade teachers separately for test scores and course grades, and find that they have systematically different impacts by student gender: teachers with high test score value-added disproportionately benefit girls' test scores and course grades, while teachers with high course grade value-added disproportionately benefit boys' test scores and course grades. I develop a two-factor model wherein unobserved cognitive and non-cognitive skills combine in different proportions to produce observed test scores and course grades, with test scores placing a higher weight on cognitive skills than grades. Under this framework, observed gender gaps imply that boys have a relative proficiency in cognitive skills and girls have a relative proficiency in non-cognitive skills. A teacher's effectiveness for a given skill dimension has larger impacts for students with a relative deficiency in that skill. In addition to predicting the gender-differentiated impacts, the framework predicts that teachers should have heterogeneous impacts connected to students' relative skill standing – independent of gender – a prediction consistent with a student-level test. The effects extend to high school: fifth-grade teachers with high course grade value-added raise boys' likelihood of remaining in the public school system, having a recorded GPA, and graduating from high school, with a smaller and statistically insignificant effect for girls, while test score value-added predicts standing in the upper part of the high school GPA distribution for boys and girls alike. These findings link multidimensional teacher effectiveness with multidimensional gender gaps in student achievement, suggesting that teacher effectiveness is not a single attribute and that its multidimensionality matters for which students benefit.

JEL Classification: I21, I24, J16.

1 Introduction

Gender gaps in educational achievement vary across subjects and assessment types. In math and science, test scores often show girls lagging behind boys in early grades, while teacher-assigned course grades tend to favor girls. In reading, both test scores and course grades favor girls, but the grade gap is larger than the score gap. In fact, gender gaps in grades exceed those in test scores for both subjects, emerge early, and grow as students progress through

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school.¹ They persist even after conditioning on test scores, suggesting that course grades capture skills beyond what test scores reflect – non-cognitive skills like persistence, organization, and engagement (among others) – that teachers recognize and reward in their evaluations of student performance (Cornwell, Mustard, and Van Parys, 2013).²

A growing body of research speaks to the importance of gender gaps in non-cognitive skills – not least because these gaps help explain gender differences in educational attainment (Aucejo and James, 2019; Autor et al., 2019; Jacob, 2002).³ There is also growing evidence that teachers have persistent impacts on test scores (Rivkin, Hanushek, and Kain, 2005; Chetty, Friedman, and Rockoff, 2014b) and course grades (Jackson, 2018; Petek and Pope, 2023), and that these two sets of impacts are distinct from one another – which the literature has interpreted as effects on cognitive and non-cognitive skills respectively. Yet these two literatures have developed largely in parallel, leaving a natural question unanswered: do teachers have systematically different impacts on boys and girls, depending on which skills they develop most effectively?

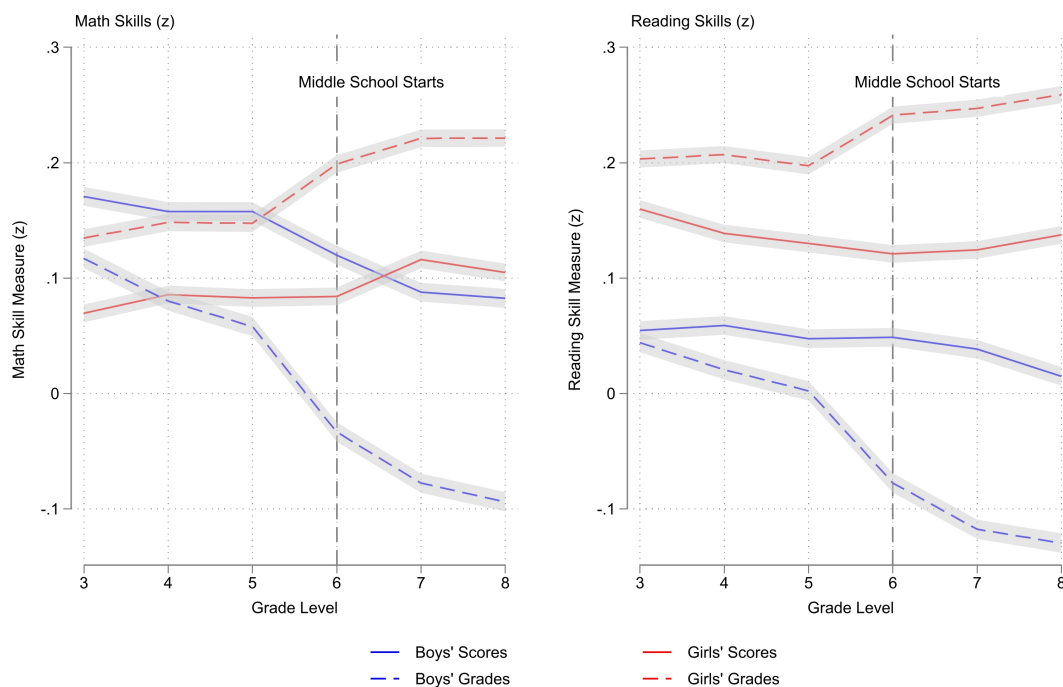
To examine this question, I use longitudinal administrative data from North Carolina, focusing on fifth-grade teachers and the middle school outcomes of their students. I begin by establishing that the descriptive patterns documented in the literature are evident in this sample as well. Figure 1 illustrates the evolution of gender gaps between third and eighth grade for three cohorts of students. Gender gaps in course grades (dashed lines) grow substantially as students progress through elementary and middle school, ultimately exceeding those in test scores (solid lines) for both math and reading. These gaps persist (and indeed, expand) even after conditioning on test scores (Appendix Figure A.1). Section 2 documents these patterns in greater detail.

¹These patterns are documented across multiple settings; see Fryer and Levitt (2010), Penner and Paret (2008), Reardon et al. (2019), Cornwell, Mustard, and Van Parys (2013), Terrier (2020), and Fortin, Oreopoulos, and Phipps (2015), among others.

²The phrase “non-cognitive skills” has been used in a variety of contexts across economics and psychology, ranging from personality traits and socio-emotional abilities to observed behaviors that predict outcomes beyond more cognitive measures of ability such as standardized or IQ tests (Heckman, Stixrud, and Urzua, 2006; Heckman, Pinto, and Savelyev, 2013; Heckman and Kautz, 2013; Lindqvist and Vestman, 2011). In the education literature, the term has also been applied to skills inferred from observed behaviors such as attendance, disciplinary outcomes, or classroom engagement, which may influence teacher evaluations and long-term outcomes (Jackson, 2018; Petek and Pope, 2023). My usage follows Cornwell, Mustard, and Van Parys (2013), who interpret course grades as reflecting skills not captured by test scores. While the term itself is imperfect, in this paper I interpret it as capturing non-test-based dimensions of academic performance.

³By the end of high school, boys have higher dropout rates and lower graduation rates than girls, and are also more likely to face disciplinary actions and less likely to attend college. Dropout rates in 2021: 6.1% males vs. 4.2% females (National Center for Education Statistics, 2023); on-time graduation rates in 2022-23: 84.9% vs. 89.9% (National Center for Education Statistics, 2024); suspensions/expulsions in 2020-21: boys accounted for the majority across grade levels (U.S. Department of Education, Office for Civil Rights, 2023); college enrollment in Oct. 2024: 55.4% males vs. 69.5% females among high school graduates ages 16-24 (U.S. Bureau of Labor Statistics, 2025).

Figure 1: Math and Reading Skills: Evolution of Gender Gaps
Third-Eighth Grade



Note. Test scores and course grades are standardized within grade-year. The sample is restricted to students who can be traced from third-eighth grade. I describe more details about the sample restrictions in [Section 2.1](#).

Next, focusing on fifth-grade teachers, I estimate teacher value-added measures separately for test scores and course grades.⁴ The two measures capture distinct dimensions of teacher effectiveness: in my data, test score value-added and course grade value-added are only weakly correlated, indicating that the teachers most effective at raising test scores are largely not the teachers most effective at raising course grades. Finally, I examine how these two dimensions of teacher effectiveness affect downstream outcomes, allowing for the impacts to differ between boys and girls.

My main finding is that teachers impact boys and girls differently, depending on which skills they improve most. A one-unit increase in fifth-grade teachers' course grade value-added (measured in standard deviations of course grades) improves boys' middle school grades by about 0.20 to 0.25 standard deviations, compared to roughly 0.10 standard deviations for girls – a difference of about 0.15 standard deviations, which is statistically significant. The effect is strongest in reading grades, with a directionally similar but less precisely estimated effect in math. Teachers with high course grade value-added also raise boys' test scores more than girls', though the cross-effect is concentrated in math: boys' math test scores increase by 0.10 to 0.12 standard deviations, with no significant effect for

⁴Existing work on multidimensional teacher impacts often aggregates course grades with absences, suspensions, and grade retention into a composite behavioral index (Jackson, 2018; Petek and Pope, 2023). I estimate value-added measures for each component separately and find that, at the elementary level, only course grade value-added passes both forecast-unbiasedness (Chetty, Friedman, and Rockoff, 2014a) and placebo tests for student sorting on prior gains (Rothstein, 2010; Rothstein, 2017). [Section 3.3](#) reports these results in greater detail.

girls; the gender differential is significant at the 10% level. The cross-effect on reading test scores is not statistically significant. Importantly, these gender-differentiated effects are estimated using value-added measures constructed on downstream grades, so they are not driven by differences in fifth-grade teachers' own grading practices.⁵

A symmetric pattern holds for test score value-added. A one-unit increase in fifth-grade teachers' test score value-added improves girls' middle school math scores by about 0.19 to 0.23 standard deviations compared to about 0.13 standard deviations for boys – with the differential effect significant in seventh and eighth grade. The estimated differential effect for reading test scores is smaller in magnitude and imprecisely estimated (girls ~0.20 vs. boys ~0.17 standard deviations), but points in the same direction. Teachers with high test score value-added also raise girls' course grades more than boys', with a differential of about 0.05 to 0.10 standard deviations across subjects – which is imprecisely estimated.⁶

To interpret these findings, I develop a simple theoretical framework, in which unobserved cognitive and non-cognitive skills combine to produce observed test scores and grades. Test scores are relatively more intensive in cognitive skills, and course grades are relatively more intensive in non-cognitive skills. This is a more flexible version of the assumption made in prior work, which treats test scores as measures of cognitive skills and the residual component of grades as a proxy for non-cognitive skills (Cornwell, Mustard, and Van Parys, 2013; Jackson, 2018; Petek and Pope, 2023). Under this framework, the observed pattern of gender gaps, wherein grade gaps exceed score gaps in both subjects, implies that boys have a relative proficiency in cognitive skills and girls have a relative proficiency in non-cognitive skills, without any implication about absolute levels of either skill.⁷ Teachers affect test scores and grades by changing the underlying cognitive and non-cognitive skills these outcomes reflect. The impact of improving a particular skill will be larger for students who are relatively deficient in that skill. The framework therefore predicts that teachers who improve cognitive skills (reflected in high test score value-added) should disproportionately benefit girls, while teachers who improve non-cognitive skills (reflected in high course grade value-added) should disproportionately benefit boys. These predictions match the empirical findings documented previously. In this framework, gender heterogeneity in teacher impacts is not a direct effect of student gender, but a consequence of how gender correlates with students' relative skill mixes.

A direct test of this interpretation asks whether teachers' impacts systematically differ according to students' relative

⁵Specifically, course grade value-added for a fifth-grade teacher measures her average contribution to her students' course grades in sixth grade. I describe the distinction between test score and course grade value-added in greater detail in [Section 3.1](#).

⁶These findings are reinforced by a complementary exercise using gender-specific value-added measures. I construct boy-specific and girl-specific value-added measures by estimating teachers' value-added separately on each gender, and test whether teachers most effective for one gender are similarly effective for the other. Across math and reading, teachers' boy-specific test score value-added predicts test score improvements for both genders while their girl-specific value-added predicts these improvements only for girls; the pattern reverses for course grades.

⁷Let c and n denote (unobserved) cognitive and non-cognitive skills respectively. Group A has a *relative proficiency* in cognitive skills (and a *relative deficiency* in non-cognitive skills) compared to group B if $\frac{c_A}{n_A} > \frac{c_B}{n_B}$; the reverse inequality implies a relative proficiency for Group A in non-cognitive skills and for Group B in cognitive skills. For more details, see [Section 4.1](#).

skill standing, regardless of gender. In [Section 5.3](#), I construct a student-level measure of relative proficiency from baseline test scores and grades, capturing whether a student performs relatively better on tests or on grades. I then examine (a) how this measure of relative skills correlates with student gender, and (b) whether teachers' impacts vary with this measure in the direction the framework predicts. Boys are disproportionately represented among students who perform relatively better on tests than on grades, and girls among students who perform relatively better on grades than on tests – consistent with the framework's implication. Furthermore, teachers with high test score value-added have larger impacts on students whose baseline performance reflects a relative deficiency in test scores, and teachers with high course grade value-added have larger impacts on students whose baseline performance reflects a relative deficiency in course grades. The measure is necessarily discrete and therefore noisy, but the patterns are directionally consistent with the framework's prediction across outcomes, even when not always precisely estimated.

These gender-differentiated impacts extend to long-run outcomes. A one-unit increase in fifth-grade teachers' course grade value-added raises boys' probability of remaining in the public school system through the start of high school by about 4.7 percentage points (which is imprecisely estimated and not significant), and their probability of remaining and graduating from a public high school within six years by about 9 percentage points (which is statistically significant); both effects are substantially smaller for girls. The two value-added measures also predict different parts of the high school GPA distribution: course grade value-added predicts whether students have a recorded transcript at all (with the effect largely driven by boys), while test score value-added predicts students' performance in the upper tail of the GPA distribution (with similar effects for boys and girls). The gender-differentiated effects of teachers thus appear to persist beyond middle school, though the long-run estimates are less precise than the elementary- and middle-school results.

This paper documents that teachers effective at raising test scores disproportionately benefit girls, while teachers effective at raising course grades disproportionately benefit boys – and that these are largely not the same teachers. It contributes to a literature documenting that gender gaps in non-cognitive skills have important consequences for educational trajectories. Jacob (2002) establishes that these gaps – in grades, efforts, and behavior – account for a substantial share of the gender gap in college enrollment, while Fortin, Oreopoulos, and Phipps (2015) document the widening of gender gaps in grades over time. A growing body of work shows that interventions targeting non-cognitive skill development have differential impacts by gender. Card, Chyn, and Giuliano (2026) find that being placed in a gifted program leads to gains in non-cognitive skills for disadvantaged boys – which in turn increases their likelihood of graduating high school and enrolling in college on time. Heckman and Karapakula (2019) document

that long-run treatment effects for males in the Perry Preschool Program are larger than those for females.⁸ While these studies evaluate targeted interventions, this paper extends this body of work by identifying which teachers in everyday classrooms produce the differential effects that benefit boys. The teachers whose effectiveness reaches boys most strongly are those who develop the non-cognitive skills that course grades reflect – not those identified by test score value-added. These teachers’ effects extend to high school graduation, where course grade value-added increases the probability that boys remain in the public school system and graduate high school, while test score value-added does not (Section 5.4).

This paper also speaks to two strands of literature on teacher effects. The first establishes that teachers have persistent and distinct effects across multiple dimensions of student achievement (Jackson, 2018; Petek and Pope, 2023), but has not asked which students benefit from each dimension. The second documents gender heterogeneity in teacher impacts through teacher characteristics or behavior – role-model effects (Bettinger and Long, 2005; Carrell, Page, and West, 2010; Dee, 2005) and biased evaluation (Carlana, 2019; Lavy and Megalokonomou, 2024; Lavy and Sand, 2018; Martínez, 2025; Terrier, 2020) – rather than through the structure of teachers’ multidimensional effectiveness. Two recent papers connect teachers’ gender-differentiated effects to gender-specific value-added measures, with Barrios-Fernández and Riudavets-Barcons (2024) emphasizing biased grading and García-Echalar, Poblete, and Rau (2024) emphasizing female teachers as role models. However, neither examines how these effects connect to the multidimensional structure of teacher effectiveness.⁹ I estimate teacher value-added on two outcomes (test scores and course grades) which reveals the asymmetric pattern in teachers’ gender-differentiated impacts. These results have implications beyond the literature: a category of effective teachers, whose strength lies in developing the skills that grades capture, is not visible through test score value-added alone – and their impacts are largest for students with a relative deficiency in the skills not adequately captured by test scores.

An interpretive contribution of this paper is the relative skills framework I develop in Section 4. The framework posits that teachers’ gender-differentiated impacts arise from differences in students’ baseline skill mixes, thereby offering an alternative to the two leading explanations in the literature, which attribute gender heterogeneity to teacher-student gender match (role-model effects) or teachers’ evaluation behavior (biased evaluation). Both alternatives locate the source of heterogeneity in teacher characteristics rather than in students’ baseline skills. Chetty, Friedman, and Rockoff (2014b) document that girls have higher long-run returns to high test score value-added teachers but do not offer an explanation for the heterogeneity; the framework I develop helps address that gap. The framework also predicts that teachers should have heterogeneous impacts based on students’ baseline skill mixes

⁸Evidence from the developmental psychology literature is broadly consistent with this pattern. Moffitt et al. (2011) document both that girls have higher childhood self-control than boys and that childhood self-control predicts a range of adult outcomes including health, earnings, and criminal behavior. Diamond and Lee (2011) find that children with weaker baseline executive function show the largest gains from interventions targeting these skills.

⁹Aucejo et al. (2022) also construct gender-specific value-added measures in reading and document substantial heterogeneity across boy-specific and girl-specific measures, but do not use these measures to predict outcomes for boys and girls.

independent of gender, and a student-level test of this prediction is consistent with the framework.

The remainder of this paper is organized as follows. In [Section 2](#), I describe the data, sample restrictions, and summary statistics for students and teachers. In [Section 3](#), I describe the construction of teacher value-added measures and present the results of the validity tests for these measures. In [Section 4](#), I develop a theoretical framework of relative skills that generates predictions matching the asymmetric pattern of gender-differentiated teacher impacts, and outline the empirical specifications for estimating these impacts. In [Section 5](#), I present my main findings on the persistence of teacher effects and their heterogeneous impacts by student gender, and report a student-level test of the relative-skills interpretation. Finally, [Section 6](#) concludes.

2 Data and Summary Statistics

2.1 Sample Construction

I use administrative data from the North Carolina Education Research Data Center (NCERDC). North Carolina’s data has been used in several influential studies on teacher value-added, including Jacob, Lefgren, and Sims ([2010](#)), Rothstein ([2010](#)), Rothstein ([2017](#)), Jackson ([2018](#)), and Aucejo and James ([2019](#)), among several others. The NCERDC dataset covers public school students and teachers who can be tracked longitudinally, and includes detailed academic, behavioral, and long-run outcomes that allow me to estimate teacher effects across multiple dimensions of student achievement using complete student cohorts.

The analysis spans 2006 to 2013, based on data availability. My primary source for academic outcomes (specifically, test scores and course grades) and teacher-student linkages is the End of Grade (EOG) files in the NCERDC data. Course grades for elementary and middle school students are available from 2006 to 2013, with third grade as the first consistently observed grade. North Carolina reports “anticipated course grades” at the time of EOG testing, which are anticipated by teachers for each student’s in-class performance in math and ELA, prior to finalizing report card grades. These measures are recorded before the academic year ends, and serve as proxies for final grades. While the use of anticipated rather than finalized grades introduces measurement error, Mozenter ([2019](#)) argues that this error is likely to be classical, making estimates noisier but unbiased when grades are used as outcomes. Anticipated grades are also strong predictors of high school GPA (as I show in Appendix [Figure B.1](#)), and have been used in prior work to identify variation in teacher grading patterns and its effects on students’ outcomes in later grades. Although anticipated grades are less commonly used than finalized report card grades, the setting of North Carolina elementary schools – where self-contained classrooms mean the same teacher both instructs and evaluates students in math and reading – makes them well suited to this analysis.

Students are linked to teachers via the EOG files, which serve as the primary source for academic outcomes. Teacher

identifiers in the EOG testing files are available only through 2011, establishing 2006 to 2011 as the estimation window for fourth- and fifth-grade teachers' value-added measures. Value-added (VA) measures are constructed using a restricted sample that satisfies four conditions. First, the teacher must be linked to a student in a given grade-year via the EOG file and recorded as having taught a self-contained, non-special-education elementary classroom for that grade-year in the School Activity Report (SAR) file. Second, VA estimates are identified only for teachers linked to at least 12 students in a given classroom, consistent with standard practice in the literature. Third, the estimation sample is limited to teachers of students in grades 4 and 5 during 2006 to 2011; anticipated course grades are not available before 2006, and teacher identifiers for elementary school classrooms are not available after 2011 in the EOG files. Finally, to enable within-school estimation, schools must have at least two teachers per grade in a given year. In elementary grades, the same teacher typically teaches both math and reading, enabling consistent identification of teacher effects for both subjects.

The student characteristics I use include sex, race, economic disadvantage, English as a second language (ESL), and whether the student reported a disability.¹⁰ In addition to academic outcomes, I use three behavioral measures (absences, suspensions, and grade repetition) and a composite behavioral index that I construct from these measures, following the methodology of Jackson (2018).¹¹ I evaluate these behavioral measures as candidate non-cognitive value-added outcomes in [Section 3](#) and [Appendix C.1](#). The data also include three long-run outcomes for these students: weighted high school GPA (from the GPA data files), and high school dropout and graduation (from the school exit files).

The analysis sample includes students who began third grade between 2006 and 2008, which allows me to observe the same students from third grade through middle school (2011 to 2013, assuming normal progression). This design lets me investigate the impacts of elementary teachers on middle school outcomes while preserving the student-teacher linkages required to estimate value-added measures. The resulting analysis sample consists of the 111,475 students who can be traced from third through eighth grade and satisfy the sample conditions described above.¹²

Appendix [Table B.1](#) reports summary statistics for the unrestricted sample of traceable students, alongside differences between students included in and excluded from the analysis sample. Students in the analysis sample are

¹⁰Economically disadvantaged students are defined as those eligible for a free or reduced price lunch.

¹¹I construct two versions of the composite index following Jackson (2018); one that combines course grades with absences, suspensions, and grade repetition, and one that combines only the three behavioral measures. The summary statistics report the latter, since course grades are reported separately.

¹²Requiring students to be traceable through eighth grade technically conditions on a post-treatment administrative outcome, which could introduce selection bias if fifth-grade teachers substantially alter students' likelihood of remaining in the state data. However, the potential for teacher-induced middle-school attrition is small in this setting. As documented in Appendix [Table B.2](#), middle-school grade repetition is rare (roughly 0.4 percent), and Appendix [Table C.2](#) shows that teacher value-added for grade repetition does not reliably predict later retention. I therefore proceed under the assumption that routine middle-school traceability is largely independent of fifth-grade teacher value-added.

positively selected relative to the full set of traceable students: at the end of fourth grade, they have higher test scores and course grades, and a stronger behavioral profile. The analysis sample also contains fewer economically disadvantaged, ESL, and disabled students, fewer Black, Hispanic, and Asian students, and somewhat larger classes, though these demographic differences are modest.

2.2 Summary Statistics: Students

[Table 1](#) reports summary statistics for the 111,475 students in the analysis sample. Panel A summarizes student demographics and class size. Panel B reports test scores and course grades in math and reading, each standardized within grade-year and pooled separately for elementary (third through fifth grade) and middle school (sixth through eighth grade). Panel C reports long-run outcomes: whether a student appears in the graduation-cohort sample, as well as jointly-defined indicators for appearing in the graduation-cohort sample, and graduating within four and six years and for having a recorded high school GPA.

Two patterns are worth highlighting here. First, the gender gap in course grades is substantially larger than the gap in test scores, and it widens as students move from elementary to middle school (see [Figure 1](#)). In reading test scores, girls outperform boys by 0.08 to 0.12 standard deviations at every grade level. In math test scores, boys begin third grade with a 0.1 standard deviation advantage, but this advantage shrinks and reverses by seventh grade. In course grades, girls outperform boys in both math and reading, with the gaps growing from roughly 0.1 standard deviations in third grade to 0.4 standard deviations by eighth grade. The reading grade gap, in particular, roughly doubles between elementary and middle school, while the gap in reading test scores remains stable. Second, these gender differences extend to long-run outcomes. Observing high school graduation requires that a student appear in North Carolina's cohort graduation rate (CGR) sample; roughly 91 percent of the analysis sample does so. Boys are slightly less likely than girls to appear in this sample. The remaining outcomes are defined jointly with sample membership, taking the value one only if a student appears in the CGR sample and achieves the outcome: boys are correspondingly less likely to be recorded as graduating within four or six years, and less likely to have a recorded high school GPA. I discuss the construction of these joint outcomes, and the selection concern that motivates it, in [Section 5.4](#). Boys also have weaker behavioral profiles than girls, with more absences and higher rates of suspension and grade retention; these gaps expand as students progress into middle school ([Appendix Table B.2](#)).

Demographic differences between boys and girls in the analysis sample are modest. Girls are slightly more likely to be classified as economically disadvantaged, while boys are marginally more likely to be White. This pattern is partly explained by differential selection into the analysis sample. Disadvantaged and Black students are less likely to be selected into the analysis sample, as I show in [Appendix Table B.1](#). Since 72 percent of Black students are classified as disadvantaged, compared to 37 percent of non-Black students, the exclusions disproportionately affect

disadvantaged Black boys, leaving a relatively higher share of White boys, corroborating the findings of Autor et al. (2019). In contrast, disadvantaged girls are more likely than their male counterparts with the same characteristics to persist in the sample, which helps explain why girls in the analysis sample are somewhat more likely to be classified as disadvantaged.

2.3 Summary Statistics: Teachers

I obtain the data for teachers' observable characteristics from the personnel files in the School Activity Reports. Appendix Table B.3 reports summary statistics for teachers' gender, race, education and experience for the full set of eligible teachers, and teachers in the analysis sample of students. The analysis sample of teachers closely resembles the full set in terms of observable characteristics. Roughly 87-88 percent of teachers are female. About 85 percent are White, 14 percent are Black, with very few Hispanic teachers. Around one-third of teachers hold a graduate degree. Teachers in the analysis sample are also slightly more experienced on average (10.1 years versus 9.1 years).

3 Teacher Value-Added Measures

For estimating teacher value-added measures, I follow an approach adapted from Kane and Staiger (2008), Jackson (2018), and Petek and Pope (2023), with refinements from more recent literature that I describe below.

3.1 Estimating Teacher Value-Added Measures

I begin by estimating teacher value-added with test scores as outcomes. For a student i in classroom c , school s , and cohort t , I model the grade- g test score as

$$y_{icst}^g = \beta_0 + \mathbf{Y}_{icst}^{g-1} \beta_1 + \mathbf{X}_i \beta_2 + \gamma_{cst}^{g-1} + \phi_s^g + \epsilon_{icst}^g \quad (1)$$

Here, y_{icst}^g is the test score of student i in grade g , $\mathbf{Y}_{icst}^{g-1}$ is the vector of lagged individual, classroom, and school outcomes from grade $g - 1$, $\mathbf{X}_i$ is a vector of student demographics, γ_{cst}^{g-1} denotes fixed effects for student i 's classroom in grade $g - 1$ (thereby accounting for the role of prior teachers in any potential sorting, as described by Gilraine and McCarthy (2024) and Staiger, Kane, and Johnson (2025)), and ϕ_s^g denotes fixed effects for student i 's school in grade g .¹³ I estimate value-added separately for math, reading, and the average of the two subjects, for both test scores and course grades, yielding three test score measures and three course grade measures. The composition of $\mathbf{Y}_{icst}^{g-1}$ matches the subject granularity of the outcome: subject-specific value-added (math or reading)

¹³I define a classroom in elementary school as a school-grade-teacher-cohort tuple.

Table 1: Summary Statistics: Analysis Sample

<i>Panel A: Demographics</i>					
	Mean	Diff		Mean	Diff
White	0.584 (0.493)	0.018*** (0.003)	Disadvantaged	0.457 (0.498)	-0.018*** (0.003)
Black	0.243 (0.429)	-0.016*** (0.003)	ESL	0.055 (0.228)	0.008*** (0.001)
Hispanic	0.097 (0.296)	0.003 (0.002)	Reported Disability	0.085 (0.279)	0.059*** (0.002)
Asian	0.020 (0.142)	-0.001 (0.001)	Class Size (3rd)	19.759 (3.089)	0.027 (0.019)
Other	0.056 (0.230)	-0.004** (0.001)	Class Size (4th)	21.807 (3.436)	0.047* (0.021)
N = 111,475					
<i>Panel B: Academic Outcomes</i>					
	Elementary School (3rd-5th Grade)			Middle School (6th-8th Grade)	
	Mean	Diff		Mean	Diff
Test Score (z)	0.109 (0.875)	-0.003 (0.003)		0.091 (0.884)	-0.049*** (0.003)
Math (z)	0.120 (0.947)	0.083*** (0.003)		0.099 (0.958)	-0.005 (0.003)
Reading (z)	0.099 (0.943)	-0.089*** (0.003)		0.082 (0.949)	-0.094*** (0.003)
Course Grade (z)	0.115 (0.893)	-0.119*** (0.003)		0.076 (0.879)	-0.320*** (0.003)
Math (z)	0.115 (0.953)	-0.058*** (0.003)		0.077 (0.973)	-0.282*** (0.003)
Reading (z)	0.115 (0.946)	-0.180*** (0.003)		0.075 (0.971)	-0.357*** (0.003)
N = 111,475					
<i>Panel C: Long-Run Outcomes</i>					
	Mean	Diff		Mean	Diff
In CGR Sample	0.909 (0.288)	-0.008*** (0.002)	Graduated within 6 (Joint)	0.847 (0.360)	-0.025*** (0.003)
Graduated within 4 (Joint)	0.820 (0.384)	-0.026*** (0.003)	Has Recorded GPA (Joint)	0.761 (0.427)	-0.031*** (0.003)
N = 111,475					

Note: Reported means are for the analysis sample, with standard deviations in parentheses below. Differences are computed as the boys' mean minus the girls' mean in the analysis sample, with standard errors of the difference in parentheses below. Means and differences in Panel B are computed at the student-grade level, pooled within elementary and middle school respectively. In Panel C, "In CGR Sample" is an indicator for appearing in North Carolina's cohort graduation rate (CGR) sample; the remaining outcomes are jointly defined, taking the value one if a student appears in the CGR sample *and* achieves the outcome, and zero otherwise. This construction, and the reason for it, is discussed in [Section 5.4](#). Test scores and course grades are standardized within grade-year (z-scores). Stars denote significance levels for a t-test for differences in means: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

is estimated conditioning on subject-specific lagged test scores and course grades, while subject-averaged value-added is estimated conditioning on lagged test scores and course grades averaged across subjects. In addition to lagged academic outcomes, $\mathbf{Y}_{icst}^{g-1}$ includes absences and indicators for grade repetition and suspensions for student i in grade $g - 1$, following the multidimensional value-added framework laid out by Jackson (2018). $\mathbf{Y}_{icst}^{g-1}$ also includes leave-one-out classroom and school-level averages of course grades, test scores, absences, suspensions, and grade retention, to account for the potential to be assigned to a given teacher based on prior-grade peer characteristics (Clotfelter, Ladd, and Vigdor, 2006; Horváth, 2015). All components of $\mathbf{Y}_{icst}^{g-1}$ enter the model as cubic polynomials to account for non-linearities in how these outcomes grow, following standard practice in the literature. The vector $\mathbf{X}_i$ includes student i 's race, gender, whether they had english as a second language, whether they were economically disadvantaged, and whether they reported a disability. The residual ϵ_{icst}^g is the component of achievement not predicted by prior outcomes, demographics, or peer composition.

To estimate value-added measures for outcomes other than test scores, I modify this setup by using the lead of the outcome (i.e., y_{icst}^{g+1}), mirroring the approach followed by Jackson (2018) and Petek and Pope (2023). While this introduces more noise in the model (since it partially captures the effect of the teacher in grade $g + 1$ as well), it reduces any potential biases stemming from grading practices differing between teachers, or from teachers evaluating their own students. Specifically, sixth-grade course grades are regressed on fourth-grade outcomes and covariates to form value-added measures for fifth-grade teachers. The right-hand-side controls mirror those in equation (1), with lagged subject-specific test scores and course grades included in $\mathbf{Y}_{icst}^{g-1}$. Value-added measures for behavioral outcomes are constructed analogously (using lead outcomes), which is described in greater detail in [Appendix C.1](#).¹⁴

Following Kane and Staiger (2008), Koedel, Mihaly, and Rockoff (2015), and Petek and Pope (2023), the residual ϵ_{icst}^g can be additively decomposed into a teacher value-added term δ_j^g , a classroom-specific term μ_c^g , and a student-specific idiosyncratic error term ν_{ist}^g .¹⁵

$$\epsilon_{icst}^g = \delta_j^g + \mu_c^g + \nu_{ist}^g \quad (2)$$

I estimate leave-cohort-out versions of [Equation \(1\)](#), and extract the residuals $\hat{\epsilon}_{icst}$ from these regressions. I then average the residuals for each teacher j :

¹⁴Using sixth-grade outcomes means that the value-added measures partially captures the downstream teacher's influence. Treating this downstream component as noise requires assuming that, conditional on the model's controls, fifth-grade teachers do not systematically sort students into sixth-grade classrooms with predictably different grading standards or gender-specific grading biases. While unobserved tracking remains a potential risk, the inclusion of school and prior-classroom fixed effects, along with rich baseline controls, limits the scope for systematic sorting. In addition, I find that course grade value-added does not predict subsequent middle-school peer ability or advanced-track placement, mitigating the concern that these measures are driven by downstream track assignment (results available upon request).

¹⁵Chetty, Friedman, and Rockoff (2014a) and other literature allow for year-to-year fluctuations in teacher value-added, by including a drift term in the model. I refrain from this, since my estimations are based on fewer eligible cohorts.

$$\widehat{\delta}_j^g = \frac{1}{n_j^g} \sum_{i \in j} \widehat{\epsilon}_{icst}^g \quad (3)$$

$\widehat{\delta}_j^g$ provides an unbiased estimate of δ_j^g as long as $\mathbb{E}[\mu_c^g + \nu_{icst}^g | j] = \mathbb{E}[\mu_c^g + \nu_{icst}^g]$, i.e., conditional on the controls specified in Equation (1), teachers don't receive students that systematically differ on unobserved achievement. Chetty, Friedman, and Rockoff (2014a) and Bacher-Hicks, Kane, and Staiger (2014) show that the controls used for students' prior achievement and behavior account for most student sorting in value-added models, and Gilraine and McCarthy (2024) and Staiger, Kane, and Johnson (2025) show that the inclusion of prior teacher fixed effects (captured through γ_{cst}^{g-1} in Equation (1)) further absorbs unobserved variation in student achievement that could otherwise generate bias through sorting. To alleviate concerns about whether students differentially sort into fifth-grade classrooms taught by better teachers, I perform a placebo test wherein I estimate the effects of high value-added fifth-grade teachers on fourth-grade outcomes, following Rothstein (2010) and Rothstein (2017). I show in Figure 3 that conditional on a third-grade baseline, fifth-grade teacher value-added does not predict fourth-grade outcomes for test scores and course grades. Finally, I use empirical bayesian methods to shrink the teacher effects towards the mean of the distribution, adapting the approach of Chetty, Friedman, and Rockoff (2014a) and Angrist et al. (2017), as shown below.

$$\widehat{\delta}_j^{EB} = \underbrace{\left(\frac{\widehat{\sigma}_\delta^2}{\widehat{\sigma}_\delta^2 + \widehat{s}_j^2} \right)}_{1-a_j} \widehat{\delta}_j + \underbrace{\left(\frac{\widehat{s}_j^2}{\widehat{\sigma}_\delta^2 + \widehat{s}_j^2} \right)}_{a_j} \bar{\delta} \quad (4)$$

Here, a_j is the shrinkage weight associated with teacher j ; higher values of a_j imply a greater amount of shrinkage towards the common mean (i.e., $\bar{\delta}$) for teacher j , due to noisier estimates of $\widehat{\delta}_j$. s_j is the standard error of $\widehat{\delta}_j$, and $\widehat{\sigma}_\delta^2 = \frac{1}{J} \sum_{j=1}^J [(\widehat{\delta}_j - \bar{\delta})^2 - \widehat{s}_j^2]$. $\widehat{s}_j^2$ is subtracted as a bias-correction term that accounts for the excess variance in $\widehat{\delta}_j$ due to sampling error (Angrist et al., 2017).

3.2 Descriptive and Summary Statistics: Value-Added Measures

Table 2 reports the standard deviations and correlations of the value-added measures for academic outcomes. I also construct value-added measures for behavioral outcomes (absences, suspensions, grade repetition, and the composite behavioral indices), which I evaluate as candidate non-cognitive measures in Section 3.3 and describe in Appendix C.1.

Panel A of Table 2 reports standard deviations.¹⁶ Test score value-added has a standard deviation of 0.104 over-

¹⁶I report the standard deviations of the unshrunk value-added measures, in order to facilitate comparisons with standard deviations reported in other literature.

Table 2: Standard Deviations and Correlations of Academic Value-Added Measures

<i>Panel A: Standard Deviations</i>				
	SD			SD
Test Score (z)	0.104		Course Grade (z)	0.132
Math (z)	0.135		Math (z)	0.153
Reading (z)	0.115		Reading (z)	0.154
<i>Panel B: Correlations, Subject-Specific Measures</i>				
	Math Score	Reading Score	Math Grade	Reading Grade
Math Score (z)	1			
Reading Score (z)	0.407	1		
Math Grade (z)	0.212	0.102	1	
Reading Grade (z)	0.102	0.149	0.449	1
Correlation between Test Score VA and Course Grade VA: 0.186				

Note: Panel A reports standard deviations of fifth-grade teacher value-added measures, expressed in standard deviations of student outcomes. Standard deviations are reported for the unshrunk value-added measures, to facilitate comparison with the literature. Panel B reports correlations among the subject-specific value-added measures. All value-added measures are estimated as described in [Section 3.1](#).

all, 0.135 for math, and 0.115 for reading – magnitudes that fall within the range documented in the literature on teacher effects on test scores (e.g., Hanushek and Rivkin, 2010; Koedel, Mihaly, and Rockoff, 2015). Course grade value-added has a standard deviation of 0.132 overall, 0.153 for math, and 0.154 for reading. Panel B reports correlations among the subject-specific measures. Within an assessment type, math and reading value-added are moderately correlated: the correlation is 0.407 between math and reading test scores (similar to the estimate obtained by Goldhaber, Cowan, and Walch (2013)) and 0.449 between math and reading course grades. Correlations across assessment types are considerably smaller. The correlation between test score value-added and course grade value-added is 0.186 (Petek and Pope (2023) report similar correlations). The same pattern holds within subjects: the correlation between math test scores and math grades is 0.212, and between reading test scores and reading grades, 0.149 – thus lending support to the argument that the two measures capture distinct dimensions of teacher effectiveness.

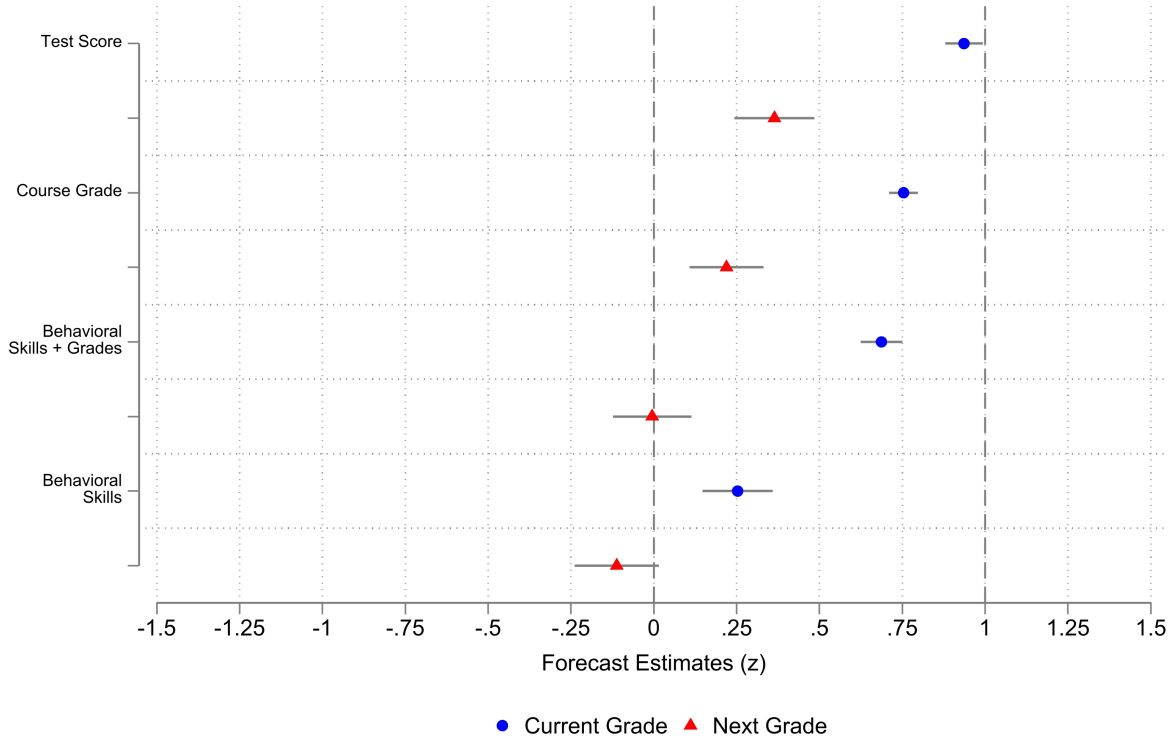
3.3 Validity Tests

To assess the validity of the estimated teacher value-added measures, I follow the approach developed by Chetty, Friedman, and Rockoff (2014a) and test whether these measures are forecast-unbiased. Intuitively, a teacher who improves student achievement by one standard deviation should, on average, generate an improvement of one standard deviation in student outcomes. Formally, I estimate the following equation:

$$y_{icst}^h = \alpha_0 + \alpha_1 \widehat{\delta}_j^g + \mathbf{Y}_{icst}^{g-1} \alpha_3 + \mathbf{X}_i \alpha_4 + \gamma_{cst}^{g-1} + \phi_s^g + u_{icst}^h \quad (5)$$

where y_{icst}^h is student i 's outcome in grade h , $\widehat{\delta}_j^g$ is the leave-cohort out empirical Bayes estimate of the value-added measure for student i 's grade- g teacher for outcome y , and all other variables are as defined in [Section 3.1](#). I construct value-added on both contemporaneous and lead outcomes, and test each against the matching outcome: contemporaneous value-added (g) against the grade- g outcome ($h = g$), and lead value-added ($g + 1$) against the grade- $(g + 1)$ outcome ($h = g + 1$). Following Chetty, Friedman, and Rockoff (2014a), a measure is *forecast-unbiased* if $\widehat{\alpha}_1 = 1$. I report the estimates of $\widehat{\alpha}_1$ for fifth-grade teachers' value-added measures based on test scores, course grades, and two composite indices for behavioral outcomes in [Figure 2](#). Results from forecast-unbiasedness tests for all value-added measures are reported in [Appendix Figure C.1](#).

Figure 2: Forecast-Unbiasedness Test for Value-Added Measures
Fifth-Grade Teachers



Note. Figure reports estimated values of $\widehat{\alpha}_1$ for value-added measures constructed for test scores, course grades, and behavioral outcomes, measured in standard deviation units of the outcomes. Estimated coefficients are from regressions wherein fifth (current) and sixth (next) grade outcomes are regressed on fifth-grade VA measures.

The contemporaneous estimates are closest to one for test score value-added, somewhat smaller for course grade value-added, and smaller still for the behavioral measures. Value-added constructed on lead outcomes – as for course grades and behavioral measures – is noisier, since it partially reflects the grade- $(g + 1)$ teacher and classroom in addition to the grade- g teacher. Test scores are also inherently less noisy than course grades: test scores are

continuous and graded against common standards within a grade-year, whereas course grades are discrete and teacher-assigned. Consistent with this, the standard errors of the unshrunk value-added measures are uniformly higher for lead outcomes than for contemporaneous outcomes, and higher for course grades and behavioral measures than for test scores, as I show in [Appendix C.1](#).¹⁷ For every outcome other than test scores, I use the lead value-added measure, for the reasons described in [Section 3.1](#). I report the contemporaneous estimates here to show that the attenuation of $\hat{\alpha}_1$ appears for every lead measure (relative to its contemporaneous counterpart), consistent with the greater degree of noise in lead outcomes. The forecast bias in $\hat{\alpha}_1$ for non-test score measures should therefore be read as attenuation bias rather than as a lack of validity: the positive, significant coefficient for course grade value-added is best interpreted as a lower bound on teachers' true impact.

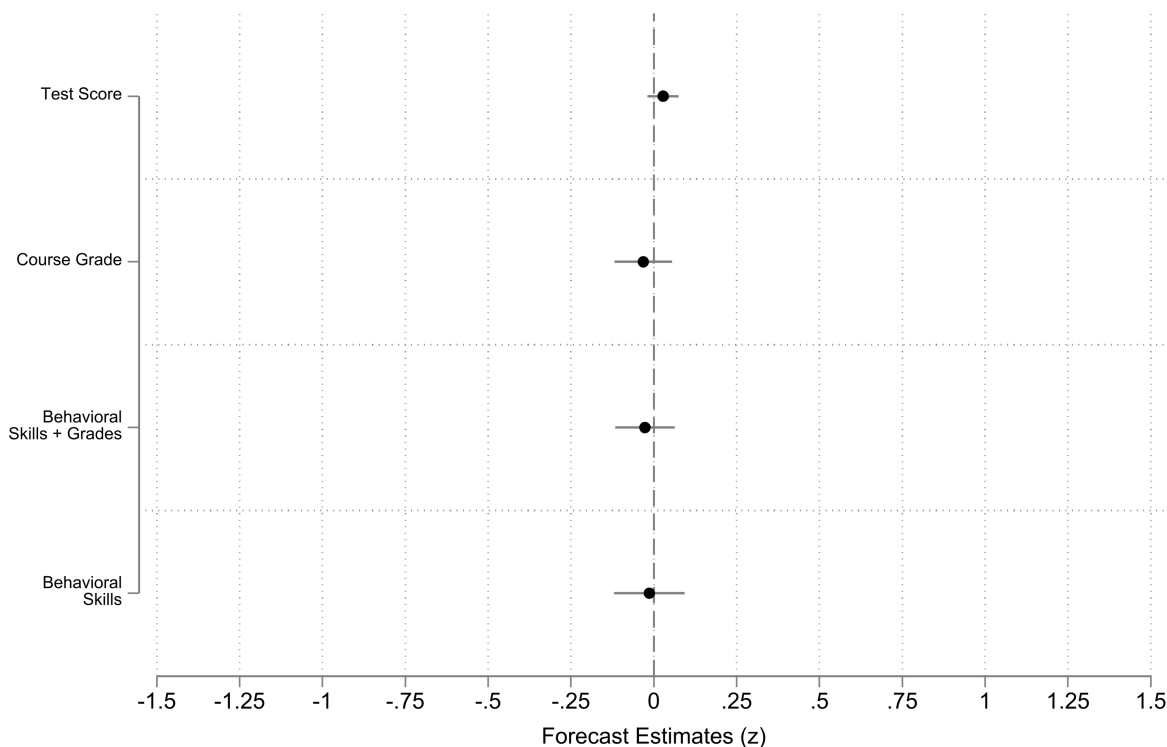
I next implement a placebo test for sorting on prior gains, following Rothstein (2010) and Rothstein (2017). I regress students' fourth-grade outcomes on the value-added measures of their fifth-grade teachers, using the specification in [Equation \(5\)](#) with $(h = g - 1)$ and with the lagged outcomes and classroom fixed effects defined for third grade $(g - 2)$.¹⁸ If the value-added measures capture teacher effectiveness rather than student sorting, a teacher's fifth-grade value-added should not predict their students' fourth-grade outcomes, and $\hat{\alpha}_1$ should be close to zero. Test score value-added and course grade value-added both pass this test: neither significantly predicts fourth-grade outcomes, as shown in [Figure 3](#).¹⁹ The two composite behavioral indices also pass the test at the composite level. However, this does not hold for the individual components of these indices. The full set of estimates for all value-added measures, reported in [Appendix Figure C.2](#), shows that value-added measures for absences significantly predict prior-grade absences, and the value-added for suspensions weakly predict prior-grade suspensions, indicating that these measures partly reflect student sorting rather than teacher effectiveness.

¹⁷I use the standard errors of the unshrunk measures since they better indicate how noisy the measures are; higher standard errors also imply greater shrinkage (see [Equation \(4\)](#)).

¹⁸I focus on fifth-grade teachers because the placebo test for student sorting on prior gains (Rothstein, 2010) requires observing students for at least two grades prior to the focal teacher, and third grade is the first year students appear in the sample. The substantive results are similar when I instead use fourth-grade teachers, and are available upon request.

¹⁹Test score value-added is defined using fifth-grade test scores; all other value-added measures are defined using sixth-grade outcomes.

Figure 3: Placebo Test for Sorting on Prior Gains
Fifth-Grade Teachers



Note. Figure reports estimated values of $\hat{\alpha}_1$ for value-added measures constructed for test scores, course grades, and behavioral outcomes, measured in standard deviation units of the outcomes. Estimated coefficients are from regressions wherein 4th grade outcomes are regressed on 5th grade VA measures.

Taken together, these results show that among the non-test score measures, only course grades yield value added measures that credibly pass the validity checks. On the forecast-unbiasedness test, course grade value-added is attenuated but positive and significant, while the two composite indices have coefficients that are not distinguishable from zero, implying that combining course grades with the behavioral components eliminates the predictive power that course grades carry on their own. On the placebo test, course grade value-added shows no sign of sorting on prior gains, whereas value-added for absences and suspensions predicts students' prior-grade outcomes. Finally, [Table C.2](#) shows that value-added for each behavioral component fails to predict that component's own outcomes later in middle school: a teacher identified as reducing absences, suspensions, or grade repetition does not measurably affect those outcomes in subsequent grades. This motivates treating course grade value-added as a standalone measure of teachers' contributions to non-cognitive skills, rather than embedding it within a composite behavioral index.

I also construct gender-specific value-added measures by re-estimating the same procedure separately on boys and girls; these are used as a supplementary check in [Section 5.2.1](#). Details on construction and results are reported in [Appendix E](#).

4 Gender-Differentiated Impacts of Teachers

4.1 The Relative Skills Framework

I model student achievement as a function of two latent skills: cognitive skills (c) and non-cognitive skills (n). These skills combine as two imperfectly substitutable inputs to produce two observable outcomes: test scores (y) and course grades (z). I operationalize this using a simple Cobb-Douglas function. The model is illustrative rather than structural; it formalizes the intuition for why teachers' effects may differ by gender.

$$y = c^\theta n^{1-\theta}, \quad \text{and} \quad z = c^\gamma n^{1-\gamma}$$

Prior work in this space has typically treated test scores as measures of cognitive skills, and the residual component of course grades (conditional on test scores) as a proxy for non-cognitive skills (Cornwell, Mustard, and Van Parys, 2013; Jackson, 2018; Petek and Pope, 2023). Implicit in this approach is that grades reflect both cognitive and non-cognitive skills, but place relatively greater weight on the latter. The model's central assumption formalizes this idea: test scores are relatively more intensive in cognitive skills and grades relatively more intensive in non-cognitive skills, so that $0 < \gamma < \theta < 1$.

Let students' gender be indexed by $\kappa \in \{B, G\}$ (boys, girls) and subjects by $s \in \{m, r\}$ (math, reading). I define observed relative outcomes for subject s between boys and girls as

$$\eta_s^y = \frac{y_{Bs}}{y_{Gs}}, \quad \text{and} \quad \eta_s^z = \frac{z_{Bs}}{z_{Gs}}$$

and I define the ratio of latent skills as

$$\psi_{\kappa s} = \frac{c_{\kappa s}}{n_{\kappa s}}$$

It follows that

$$\frac{\psi_{Bs}}{\psi_{Gs}} = \left(\frac{\eta_s^y}{\eta_s^z} \right)^{\frac{1}{\theta-\gamma}} \quad (6)$$

This expression links observed gender differences in test scores and grades to differences in the underlying relative mixes of unobserved cognitive and non-cognitive skills.

4.1.1 Empirical Gender Gaps and Relative Skills

Two stylized facts in my data guide the interpretation of gender differences in test scores and grades. At the baseline, boys have higher math test scores but lower math grades than girls. Assuming these standardized differences map to

corresponding proportional gaps on a latent positive scale, this implies that $\eta_m^y > 1$ and $\eta_m^z \leq 1$, or that $\eta_m^y/\eta_m^z > 1$. Therefore boys have a higher ratio of cognitive to non-cognitive skills than girls (i.e., $\psi_{Bm} > \psi_{Gm}$). In reading, girls outperform boys in both test scores and grades, but the grade gap is larger than the test score gap. Extending the earlier assumption made about proportional gaps, this implies that $\eta_r^y/\eta_r^z > 1$, or that boys have a higher ratio of cognitive to non-cognitive skills than girls (i.e., $\psi_{Br} > \psi_{Gr}$). Taken together, these stylized facts imply that $\psi_{Bs} > \psi_{Gs}$ for both subjects, and the differences in relative skills for boys and girls can be interpreted as boys having a relative proficiency in cognitive skills, and girls having a relative proficiency in non-cognitive skills for both subjects. It is important to note here that this interpretation of relative proficiency does not make any assumptions about which gender has greater (or lower) absolute levels of cognitive or non-cognitive skills.

4.1.2 Teachers' Heterogeneous Effects by Students' Relative Skills

Teachers who are effective at improving a particular skill disproportionately improve outcomes of students who are relatively deficient in that skill. Modeling a teacher's impact as a uniform additive shock to the relevant latent skill, a teacher who improves cognitive skills (c) will therefore improve girls' outcomes more than boys', holding fixed her influence on non-cognitive skills (n). When girls receive a teacher who improves cognitive skills in subject s , their test scores improve by $\theta \left(\frac{1}{\psi_{Gs}}\right)^{1-\theta}$, while boys' test scores improve by $\theta \left(\frac{1}{\psi_{Bs}}\right)^{1-\theta}$.²⁰ Similarly, girls' grades improve by $\gamma \left(\frac{1}{\psi_{Gs}}\right)^{1-\gamma}$ and boys' grades improve by $\gamma \left(\frac{1}{\psi_{Bs}}\right)^{1-\gamma}$. Given girls' relative deficiency in cognitive skills (i.e., $\psi_{Gs} < \psi_{Bs}$), this implies that we should expect teachers who improve cognitive skills to have larger impacts on girls' outcomes (both test scores and grades) relative to boys' outcomes.

Conversely, a teacher who improves non-cognitive skills (n) will improve boys' outcomes more than girls', holding fixed her influence on cognitive skills. When boys receive a teacher who improves non-cognitive skills in subject s , their test scores improve by $(1 - \theta) \psi_{Bs}^\theta$, while girls' test scores improve by $(1 - \theta) \psi_{Gs}^\theta$. Similarly, boys' grades improve by $(1 - \gamma) \psi_{Bs}^\gamma$, while girls' grades improve by $(1 - \gamma) \psi_{Gs}^\gamma$. Given boys' relative deficiency in non-cognitive skills (i.e., $\frac{1}{\psi_{Bs}} < \frac{1}{\psi_{Gs}}$ or $\psi_{Bs} > \psi_{Gs}$), this again implies that we should expect teachers who improve non-cognitive skills to have larger impacts on boys' outcomes (both test scores and grades) relative to girls' outcomes.

These predictions apply to both math and reading, even though the direction of the raw gaps differs across subjects. In math, boys' relative proficiency in cognitive skills and girls' relative proficiency in non-cognitive skills arise from boys' higher test scores alongside lower grades at the baseline. In reading, girls outperform boys on both outcomes, but the larger grade gap implies the same ordering of relative skills ($\psi_{Bs} > \psi_{Gs}$) in both subjects. The prediction therefore follows from the relative skill mix, not from the sign of any single raw gap: in both subjects, girls benefit more from teachers who improve cognitive skills, and boys benefit more from teachers who improve non-cognitive

²⁰I formally derive these expressions in [Appendix D](#).

skills.

4.2 Empirical Design

The predictions in [Section 4.1.2](#) concern teachers' contributions to latent cognitive and non-cognitive skills, which are not directly observed. To take them to the data, I interpret teachers' test score value-added and course grade value-added as proxies for their impacts on cognitive skills and non-cognitive skills respectively – when conditioned on each other. This is the value-added analogue of the assumption in [Section 4.1](#) that test scores are relatively more intensive in cognitive skills and grades in non-cognitive skills, and the standard convention in the literature on multidimensional teacher effects (Jackson, 2018; Petek and Pope, 2023). This is the model's key interpretive assumption, which guides the directional predictions that I take to the data. Given this conditional mapping, the predictions of [Section 4.1.1](#) imply that teachers with high test score value-added should have stronger impacts on girls, and teachers with high course grade value-added should have stronger impacts on boys, for both test score and grade outcomes.

To test for heterogeneous impacts of teachers on boys and girls, I interact teacher value-added measures with an indicator for whether the student is female. I estimate a multidimensional version of [Equation \(5\)](#), as shown below:

$$y_{icst}^h = \alpha_0 + \alpha_1 F_i + \delta_j^g \alpha_2 + (\delta_j^g \times F_i) \alpha_3 + \mathbf{Y}_{icst}^{g-1} \alpha_4 + (F_i \times \mathbf{Y}_{icst}^{g-1}) \alpha_5 + \mathbf{X}_i \alpha_6 + (F_i \times \mathbf{X}_i) \alpha_7 + \gamma_{cst}^{g-1} + \phi_s^g + u_{icst}^h \quad (7)$$

Here F_i is an indicator for whether student i is female, $\delta_j^g = [\hat{\delta}_j^{g, \text{score}} \ \hat{\delta}_j^{g, \text{grade}}]$ is the vector of teacher value-added in test scores and course grades for teacher j in grade g , and $\mathbf{Y}_{icst}^{g-1}$ and $\mathbf{X}_i$ are the controls as specified in [Equation \(5\)](#).²¹ I interact every control with F_i , so that the relationship between prior achievement, demographics, and current outcomes is allowed to differ for boys and girls. This ensures that the interaction of value-added measures with student gender captures the differential effect of teacher value-added by gender, rather than gender differences in how prior outcomes and characteristics map into current achievement. The coefficient vectors $\alpha_2 = [\alpha_2^{\text{score}}, \alpha_2^{\text{grade}}]$ and $\alpha_3 = [\alpha_3^{\text{score}}, \alpha_3^{\text{grade}}]$ separately capture the impacts of score and grade value-added. α_2 reflects the effects for boys (the baseline group), while α_3 captures the differential effects for girls relative to boys. A test for the predictions outlined in [Section 4.1](#) is α_3^{score} should be positive (i.e., test score value-added should have stronger impacts on girls), and α_3^{grade} should be negative (i.e., course grade value-added should have stronger impacts on boys). The signs for α_3^{score} and α_3^{grade} follow the predictions of the model for both outcome measures –

²¹ $\mathbf{X}_i$ excludes the indicator for student gender in this equation, to avoid redundancy.

test scores *and* grades.²²

5 Results

5.1 Persistence of Teacher Effects

I begin by establishing that the estimated teacher value-added measures capture meaningful and persistent impacts on student achievement, before turning to the gender-differentiated analysis in [Section 5.2](#). I estimate the specification in [Equation \(7\)](#) without gender differentiation, allowing both measures of teacher effectiveness to influence each outcome.

$$y_{icst}^h = \alpha_0 + \delta_j^g \alpha_1 + \mathbf{Y}_{icst}^{g-1} \alpha_2 + \mathbf{X}_i \alpha_3 + \gamma_{cst}^{g-1} + \phi_s^g + \zeta_h + u_{icst}^h \quad (8)$$

Here, y_{icst}^h is an outcome (test score or course grade) for student i in grade $h \geq g + 1$, and $\delta_j^g = [\hat{\delta}_j^{g, \text{score}} \ \hat{\delta}_j^{g, \text{grade}}]$ is the vector of test score and course grade value-added measures for teacher j who taught student i in grade g . The vectors $\mathbf{Y}_{icst}^{g-1}$ and $\mathbf{X}_i$, and the fixed effects γ_{cst}^{g-1} and ϕ_s^g are as specified for [Equation \(1\)](#). ζ_h denotes fixed effects for the grade level in which the outcome is measured. Both value-added measures are leave-cohort-out empirical-Bayes estimates, constructed as described in [Section 3.1](#). Test score value-added for a fifth-grade teacher is constructed using fifth-grade test scores, and course grade value-added for a fifth-grade teacher is constructed using sixth-grade course grades. Standard errors in all models are clustered at the level of the fifth-grade teacher.

Table 3: Multidimensional Impacts of Fifth-Grade Teachers on Middle School Outcomes

	Test Scores				Course Grades			
	6th-8th		7th-8th		6th-8th		7th-8th	
	I	II	III	IV	V	VI	VII	VIII
Test Score VA	0.190*** (0.022)	0.191*** (0.022)	0.147*** (0.024)	0.148*** (0.025)		0.088*** (0.031)		0.036 (0.036)
Course Grade VA		-0.007 (0.031)		-0.006 (0.034)	0.195*** (0.044)	0.170*** (0.045)	0.182*** (0.049)	0.172*** (0.051)
Constant	-0.142 (0.182)	-0.141 (0.182)	-0.055 (0.205)	-0.055 (0.205)	-0.104 (0.182)	-0.102 (0.182)	-0.020 (0.218)	-0.020 (0.218)
Mean	0.091	0.091	0.089	0.089	0.076	0.076	0.071	0.071
N	334425	334425	222950	222950	334425	334425	222950	222950
r ²	0.769	0.769	0.758	0.758	0.463	0.463	0.450	0.450

Note: Standard errors clustered at the fifth-grade teacher level. Test Score VA is defined on fifth-grade test scores. Course Grade VA is defined on sixth-grade course grades. All outcomes are standardized (z-scores). Models include fourth-grade classroom fixed effects and fifth-grade school fixed effects. Stars denote significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

²²As a complementary approach, I also estimate a specification that uses gender-specific value-added measures (boy-specific and girl-specific, constructed as described in [Section E](#)) and tests whether teachers' effectiveness for each gender of student differs from their effectiveness for the other. The full specification and results are reported in [Appendix E](#).

Table 3 reports the impacts of fifth-grade teacher value-added on middle school outcomes, averaged over subjects. Both value-added measures are significant predictors that persist into middle school. A one-unit increase in teachers' test score value-added (measured in standard deviations of test scores) improves students' middle school test scores by about 0.19 standard deviations on average (column I), an effect that is robust to controlling for the same teacher's course grade value-added (column II). Expressed in standard deviations of the distribution of teacher value-added (reported in Table 2), a one standard deviation increase in test score value-added for fifth-grade teachers raises students' middle school test scores by about 0.020 standard deviations. Similarly, a one-unit increase in teachers' course grade value-added improves students' middle school grades by about 0.19 standard deviations (column V), reduced slightly to 0.17 standard deviations when controlling for test score value-added (column VI). Rescaled, this implies that a one standard deviation increase in course grade value-added for fifth-grade teachers raises students' middle school grades by about 0.025 standard deviations.

Since course grade value-added is constructed using sixth-grade course grades, part of the effect for outcomes in sixth-eighth grades may reflect mechanical persistence of the construct for sixth-grade outcomes. To address this, I separately estimate impacts on outcomes measured in seventh and eighth grades. Both value-added measures for fifth-grade teachers continue to predict significant improvements in student achievement (columns III-IV and VII-VIII). For test scores, the effect falls from 0.19 to 0.15 standard deviations when I restrict the sample to seventh and eighth-grade outcomes (rescaled, this implies an effect of about 0.016 standard deviations), consistent with teacher effects on test scores fading out over time (Chetty, Friedman, and Rockoff, 2014a; Rothstein, 2010; Jacob, Lefgren, and Sims, 2010). The impacts of course grade value-added show no comparable fade-out: from 0.19 standard deviations for sixth-eighth grade outcomes to 0.18 for seventh-eighth grade outcomes (rescaled, this implies an effect of about 0.024 standard deviations).

I also find some limited evidence of multidimensional impacts, consistent with Jackson (2018) and Petek and Pope (2023). Conditional on course grade value-added, teachers who are more effective in raising test scores also improve their students' grades in middle school by about 0.09 standard deviations (column VI; rescaled, this implies an effect of about 0.009 standard deviations). This effect shrinks to 0.04 standard deviations and becomes insignificant when restricting to seventh and eighth-grade outcomes, again consistent with the fade-out of test score effects. By contrast, fifth-grade teachers' value-added in course grades does not predict their students' middle school test scores (columns II and IV).

Appendix Table F.1 reports results separately for math and reading. The broad patterns are similar across subjects: both test score and course grade value-added are significant predictors of their respective outcomes. For math (Panel A), a one-unit increase in fifth-grade teachers' test score value-added improves middle school math scores by about 0.17 standard deviations (column I; rescaled, this implies an effect of about 0.023 standard deviations),

and a one-unit increase in course grade value-added improves middle school math grades by about 0.19 standard deviations (column V; rescaled, this implies an effect of about 0.030 standard deviations). For reading (Panel B), the corresponding effects are 0.18 standard deviations on reading scores (column I; rescaled, 0.021 standard deviations) and 0.12 standard deviations on reading grades (column V; rescaled, 0.018 standard deviations).

The multidimensional impacts differ across subjects. For math, the multidimensional impacts are bidirectional: course grade value-added predicts students' math scores (about 0.05 standard deviations, column II), and test score value-added predicts students' math grades (about 0.08 standard deviations, column VI). Both effects are smaller and less precise when restricting outcomes to seventh and eighth grades (columns IV and VIII). For reading, the multidimensional effects are one-sided: test score value-added predicts reading grades, but course grade value-added does not predict reading scores.

5.2 Gender Differentiated Results

I investigate teachers' gender-differentiated impacts by estimating [Equation \(7\)](#) for middle school test scores and course grades. [Table 4](#) reports the results averaged across subjects. The coefficients in this table report the effect of a one-unit increase in teacher value-added, where value-added is measured in standard deviations of the corresponding student outcome. The discussion in [Section 5.1](#) rescaled these coefficients to express effects per standard deviation in the teacher value-added distribution; I do not apply that rescaling in the analyses that follow, since the differential effects for boys and girls (and not the absolute magnitudes) are the objects of interest.

Table 4: Gender-Differentiated Impacts of Fifth-Grade Teachers on Middle School Outcomes

	Test Scores (6th-8th)		Test Scores (7th-8th)		Course Grades (6th-8th)		Course Grades (7th-8th)	
	I	II	III	IV	V	VI	VII	VIII
Female	-0.042 (0.060)	-0.042 (0.060)	-0.143** (0.066)	-0.143** (0.066)	0.370*** (0.083)	0.370*** (0.083)	0.388*** (0.095)	0.385*** (0.095)
Test Score VA	0.165*** (0.033)	0.160*** (0.033)	0.107*** (0.035)	0.102*** (0.036)		0.065 (0.048)		-0.020 (0.054)
Female × Test Score VA	0.049 (0.046)	0.059 (0.047)	0.079 (0.049)	0.089* (0.050)		0.048 (0.065)		0.109 (0.074)
Course Grade VA		0.029 (0.044)		0.032 (0.049)	0.264*** (0.065)	0.246*** (0.066)	0.255*** (0.073)	0.260*** (0.075)
Female × Course Grade VA		-0.066 (0.061)		-0.068 (0.068)	-0.141 (0.087)	-0.154* (0.089)	-0.145 (0.097)	-0.176* (0.100)
Constant	-0.151 (0.182)	-0.152 (0.182)	-0.067 (0.205)	-0.068 (0.205)	-0.122 (0.181)	-0.120 (0.181)	-0.042 (0.218)	-0.041 (0.218)
Mean	0.091	0.091	0.089	0.089	0.076	0.076	0.071	0.071
N	334425	334425	222950	222950	334425	334425	222950	222950
r ²	0.769	0.769	0.759	0.759	0.465	0.465	0.452	0.452

Note: Standard errors clustered at the Fifth-grade teacher level. Test Score VA is defined on fifth-grade test scores. Course Grade VA is defined on sixth-grade course grades. All outcomes are standardized (z-scores). Models include fourth-grade classroom fixed effects and fifth-grade school fixed effects. Stars denote significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Girls benefit more than boys from teachers with higher test score value-added, while boys benefit more than girls

from teachers with higher course grade value-added. A one-unit increase in teachers' test score value-added improves middle school test scores by about 0.16 standard deviations for boys and 0.21 standard deviations for girls (column I), though the differential effect is estimated imprecisely for sixth-eighth grades. When I restrict the outcomes to seventh-eighth grades, the female differential grows to about 0.08-0.09 standard deviations, and is statistically significant when I condition on course grade value-added (column IV). Teachers' test score value-added also has larger impacts on girls' course grades, though the differential improvement of 0.05-0.11 standard deviations is imprecisely estimated. This is consistent with the prediction that teachers' cognitive-skill impacts are stronger for girls' outcomes – for both test scores and grades. By contrast, course grade value-added predicts improvements in middle school course grades of ~0.25 standard deviations for boys and ~0.1 standard deviations for girls (column VI). The differential improvement for boys' grades is ~0.15 standard deviations and is statistically significant. I observe a similar pattern when restricting outcomes to seventh-eighth grades. Teachers' course grade value-added also has stronger impacts on boys' test scores relative to girls', though the differential improvement of about 0.06 standard deviations for boys is imprecisely estimated. This reinforces the finding that teachers who improve grades have stronger impacts on boys' outcomes (both test scores and grades). Taken together, these patterns are consistent with the framework of relative skills developed in [Section 4.1](#).

[Table 5](#) reports subject-specific versions of [Equation \(7\)](#). Test score value-added has stronger impacts on girls' test scores than on boys' in both subjects. A one-unit increase in test score value-added improves boys' middle school test scores by 0.12-0.13 standard deviations in math and about 0.17 standard deviations in reading; the corresponding effects for girls are 0.19-0.20 and 0.20-0.23 standard deviations. The female differential is statistically significant for math test scores when outcomes are restricted to seventh-eighth grades, and imprecisely estimated for reading. Test score value-added also has larger impacts on girls' course grades than on boys', though the differential effects of 0.04 standard deviations (math grades) and 0.06 standard deviations (reading grades) are imprecisely estimated.

By contrast, course grade value-added has stronger impacts on boys' outcomes than on girls' in both subjects. Boys' grades through middle school increase by 0.21-0.24 standard deviations (math) and 0.18-0.19 standard deviations (reading) in response to course grade value-added; the corresponding effects for girls are smaller, about 0.11-0.14 standard deviations (math) and 0.03-0.04 standard deviations (reading). The differential for boys' grades is statistically significant in reading and imprecisely estimated in math. Course grade value-added also has stronger impacts on boys' test scores than on girls', with the differential effect of 0.13 standard deviations on math test scores

in 7th-8th grade significant at the 10% level.²³

Table 5: Gender-Differentiated Impacts of Fifth-Grade Teachers on Middle School Outcomes

<i>Panel A: Math Outcomes</i>								
	Math Scores (6th-8th)		Math Scores (7th-8th)		Math Grades (6th-8th)		Math Grades (7th-8th)	
	I	II	III	IV	V	VI	VII	VIII
Female	0.162** (0.072)	0.161** (0.072)	0.218*** (0.080)	0.217*** (0.080)	0.438*** (0.094)	0.439*** (0.094)	0.455*** (0.108)	0.454*** (0.108)
Test Score VA	0.138*** (0.027)	0.123*** (0.028)	0.089*** (0.029)	0.072** (0.030)		0.072* (0.038)		-0.008 (0.043)
Female × Test Score VA	0.059 (0.036)	0.073** (0.037)	0.084** (0.040)	0.103** (0.041)		0.013 (0.052)		0.047 (0.059)
Course Grade VA		0.102** (0.044)		0.117** (0.050)	0.240*** (0.062)	0.211*** (0.065)	0.190*** (0.069)	0.193*** (0.071)
Female × Course Grade VA		-0.099 (0.063)		-0.131* (0.069)	-0.095 (0.085)	-0.099 (0.087)	-0.093 (0.094)	-0.112 (0.097)
Constant	-0.186 (0.190)	-0.187 (0.190)	-0.073 (0.205)	-0.075 (0.205)	-0.336 (0.204)	-0.332 (0.204)	-0.383 (0.242)	-0.383 (0.242)
Mean	0.099	0.099	0.098	0.098	0.077	0.077	0.072	0.072
N	334425	334425	222950	222950	334425	334425	222950	222950
r ²	0.707	0.707	0.699	0.699	0.389	0.389	0.377	0.377
<i>Panel B: Reading Outcomes</i>								
	Reading Scores (6th-8th)		Reading Scores (7th-8th)		Reading Grades (6th-8th)		Reading Grades (7th-8th)	
	I	II	III	IV	V	VI	VII	VIII
Female	-0.250*** (0.072)	-0.250*** (0.072)	-0.511*** (0.081)	-0.511*** (0.081)	0.303*** (0.093)	0.304*** (0.093)	0.319*** (0.108)	0.318*** (0.108)
Test Score VA	0.167*** (0.051)	0.167*** (0.051)	0.107* (0.056)	0.106* (0.057)		0.089 (0.067)		0.036 (0.078)
Female × Test Score VA	0.032 (0.071)	0.035 (0.072)	0.061 (0.079)	0.064 (0.080)		0.000 (0.092)		0.061 (0.106)
Course Grade VA		-0.002 (0.047)		0.006 (0.053)	0.193*** (0.065)	0.181*** (0.066)	0.235*** (0.075)	0.230*** (0.076)
Female × Course Grade VA		-0.023 (0.067)		-0.020 (0.074)	-0.151* (0.086)	-0.151* (0.087)	-0.194* (0.100)	-0.202** (0.101)
Constant	-0.032 (0.235)	-0.032 (0.235)	0.018 (0.266)	0.017 (0.266)	0.143 (0.207)	0.144 (0.207)	0.336 (0.249)	0.337 (0.249)
Mean	0.082	0.082	0.080	0.080	0.075	0.075	0.070	0.070
N	334425	334425	222950	222950	334425	334425	222950	222950
r ²	0.675	0.675	0.664	0.664	0.387	0.387	0.381	0.381

Note: Standard errors clustered at the fifth-grade teacher level. Math and Reading Test Score VA are defined on fifth-grade test scores. Math and Reading Course Grade VA are defined on sixth-grade outcomes. All outcomes are standardized (z-scores). Models include fourth-grade classroom fixed effects and fifth-grade school fixed effects. Stars denote significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Taken together, these results align with the relative skills interpretation. Teachers' skill-specific impacts have stronger effects on students with a relative deficiency in that skill: test score value-added has larger impacts on girls' outcomes (in both subjects), and course grade value-added has larger impacts on boys' outcomes. All eight interaction coefficients in Table 4 carry the predicted sign, with the differentials most precisely estimated for test score value-added on math scores and course grade value-added on reading grades.

²³A related question may be whether fifth-grade teachers' effectiveness in test scores or course grades affects students' sorting into middle-school classrooms and tracks. I examine this by estimating an analogous specification using middle-school peer ability (based on leave-one-out prior scores and prior grades) and advanced-track enrollment as outcomes (results available upon request). test score value-added strongly predicts average peer ability in middle-school classrooms and enrollment in advanced courses, while course grade value-added has no predictive power; there is no strong evidence of gender-differentiated effects on either outcome. To the extent that grading standards may be stricter in more demanding classrooms, the estimated effects on downstream grades are likely lower bounds; this concern applies most strongly to test score value-added, but to the extent that course grade value-added also produces test score gains that drive sorting, the same logic may apply.

5.2.1 Gender-Specific Value-Added

As a supplementary check on the gender-differentiated findings in [Section 5.2](#), I estimate [Equation \(E.1\)](#), which uses value-added measures constructed separately for boys and girls (boy-specific and girl-specific value-added). The full construction, specification, and results are reported in [Appendix E](#).

The results in [Table E.1](#) and [Table E.2](#) ([Appendix E](#)) document an asymmetric cross-impact pattern. For test score value-added, the boy-specific measure predicts test score improvements for both genders, while the girl-specific measure predicts improvements primarily for girls. For course grade value-added, the pattern reverses: the girl-specific measure predicts course grade improvements for both genders, while the boy-specific measure predicts improvements primarily for boys. This pattern is observed in both math and reading, and is consistent with the gender-differentiated impacts documented in [Section 5.2](#).

5.3 A Test of the Framework

The relative skills framework developed in [Section 4.1](#) predicts the directions of the gender-differentiated teacher effects documented in [Section 5.2](#). Specifically, the framework posits that teachers improve students in the dimension in which they have a relative deficiency: teachers who improve cognitive skills have stronger impacts on students with a relative deficiency in cognitive skills, and teachers who improve non-cognitive skills have stronger impacts on students with a relative deficiency in non-cognitive skills. I now test these predictions directly, using a student-level measure of relative skill differences.

Cognitive and non-cognitive skills are not directly observable, but their outputs are. Following the framework in [Section 4.1](#), I treat test scores as more intensive in cognitive skills than course grades, so a student's relative performance on tests versus grades is informative about her relative cognitive versus non-cognitive skills. To make this concrete, consider two students: student A in the 90th percentile of test scores and the 30th percentile of course grades, and student B with the reverse profile. Student A has a relative proficiency in test scores and a relative deficiency in course grades; student B has the reverse. Consistent with the illustrative framework developed in [Section 4.1](#), I interpret this observable gap at the student level as a proxy for a difference in the underlying skill mix: student A has a higher c/n ratio than student B, or equivalently, a relative proficiency in cognitive skills and a relative deficiency in non-cognitive skills, while student B has the reverse.

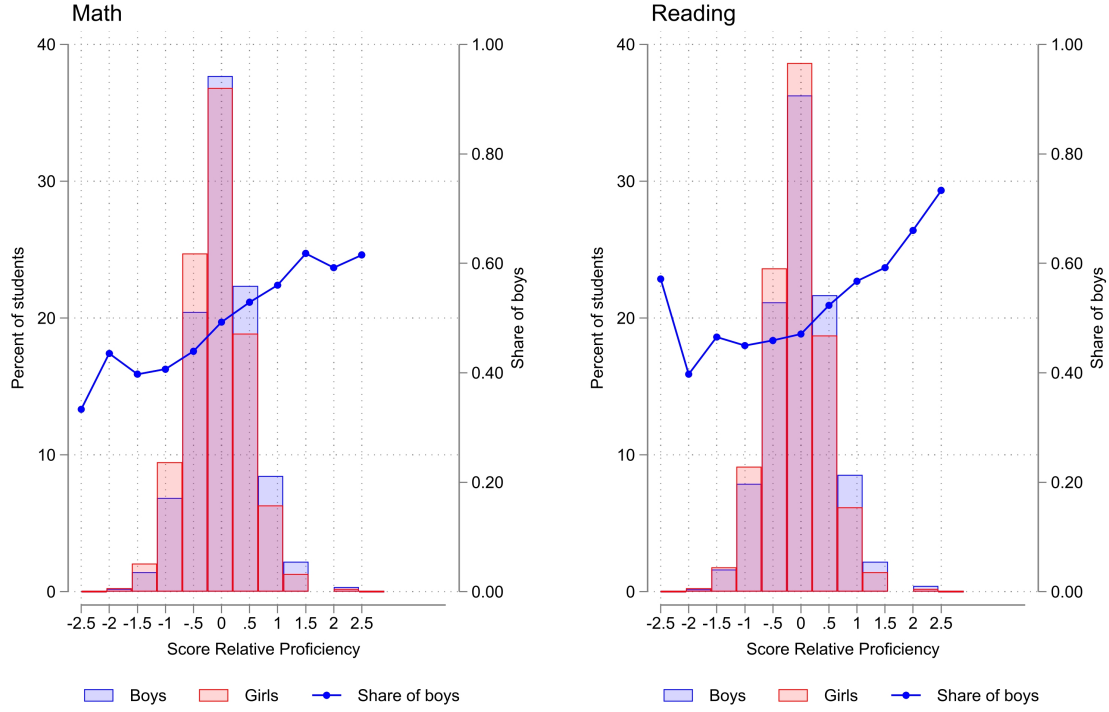
To formalize this intuition, I construct a student-level measure of relative proficiency using baseline academic performance. I take the average of third and fourth-grade test scores (y) and the average of third and fourth-grade course grades (z), separately for math and reading. For each student, I compute the test score percentile within their cohort and construct a “score-grade equivalent” (q_y) by mapping this percentile to the grade distribution. I then

define *score relative proficiency* as the difference between the score-grade equivalent and the actual grade point ($\text{Score RP} = q_y - z$).²⁴ Positive Score RP indicates a relative proficiency in test scores (equivalently, a relative deficiency in course grades); negative Score RP indicates the reverse; Score RP equal to zero indicates no relative proficiency or deficiency. The measure is discrete (taking values between -3 to $+3$ in increments of 0.5 , since course grades take four values) and necessarily noisy as a proxy for students' relative skill profiles, but is empirically usable for testing the framework's predictions.

Figure 4 plots the distributions of Score Relative Proficiency for boys and girls separately, alongside the share of boys at each value, for math (left panel) and reading (right panel). The distribution of the Score RP measure for boys' lies to the right of girls' in both subjects: the difference in mean Score RP between boys and girls is 0.11 units in math and 0.08 units in reading (both statistically significant). The share of boys increases with Score RP, as evidenced by both the histogram and the line. These patterns are consistent with the framework's prediction that boys and girls differ in their average relative skill profiles.

²⁴First, I compute within-cohort percentiles for both averaged third- and fourth-grade test scores and averaged third- and fourth-grade course grades, separately for math and reading. Because annual course grades take integer values ($0 = D$ or below, $1 = C$, $2 = B$, $3 = A$), the two-year average (z) takes on discrete values in 0.5 increments. I identify the percentile ranges corresponding to each 0.5 -point bin in the averaged-grade distribution. For example, if an average course grade of 1.5 ($z = 1.5$) spans the 11th to 20th percentile, and an average course grade of 2.0 ($z = 2.0$) spans the 21st to 40th percentile, I map test score percentiles into these same ranges to create "score-grade equivalents" (q_y). A student whose test score falls in the 11th–20th percentile receives $q_y = 1.5$, one in the 21st–40th percentile receives $q_y = 2.0$, and so on. Finally, I define the measure of score relative proficiency as $\text{Score RP} = q_y - z$. For instance, a student with a test score equivalent of $q_y = 2.0$ but an actual average course grade of $z = 1.5$ would receive a Score RP of 0.5 . I construct these measures separately for math and reading.

Figure 4: Score Relative Proficiency and Student Gender



Note. Each panel plots the distributions of Score Relative Proficiency for boys (blue) and girls (red) separately (left axis), alongside the share of boys at each value (blue line, right axis). The left panel reports math; the right panel reports reading. Score Relative Proficiency is constructed from the average of third and fourth-grade test scores and the average of third- and fourth-grade course grades, separately by subject.

I use this measure for two tests of the relative skills framework. First, a skill-heterogeneity test examines whether teacher value-added has differential effects across students with different relative skill profiles. I estimate the following model:

$$y_{icst}^h = \alpha_0 + \alpha_1 \delta_j^{VA} + \alpha_2 \text{Score RP}_i + \alpha_3 (\delta_j^{VA} \times \text{Score RP}_i) + \mathbf{Y}_{icst}^{g-1} \alpha_4 + \mathbf{X}_i \alpha_5 + \gamma_{cst}^{g-1} + \phi_s^g + \zeta_h + u_{icst}^h \quad (9)$$

Here, δ_j^{VA} denotes teacher j 's value-added measure corresponding to outcome y_{icst}^h . For simplicity, I do not account for multidimensionality in teachers' effects in this setup. When the outcome is a test score, $\delta_j^{VA} = \delta_j^{score}$, and when the outcome is a course grade, $\delta_j^{VA} = \delta_j^{grade}$. All other controls and fixed effects are as defined in Equation (8). The framework predicts $\alpha_3 < 0$ for test score outcomes, since teachers with high test score value-added have larger effects on students with lower Score RP (those with a relative deficiency in test scores). Similarly, $\alpha_3 > 0$ for course grade outcomes, since teachers with high course grade value-added have larger effects on students with higher Score RP (those with a relative deficiency in course grades).

A second test examines whether the gender-differentiated effects documented in [Section 5.2](#) reflect teachers' impacts on relative skill differences. I estimate the following model:

$$\begin{aligned}
y_{icst}^h = & \alpha_0 + \alpha_1 \delta_j^{VA} + \alpha_2 \text{Score RP}_i + \alpha_3 (\delta_j^{VA} \times \text{Score RP}_i) \\
& + \alpha_4 \text{Female}_i + \alpha_5 (\delta_j^{VA} \times \text{Female}_i) + \mathbf{Y}_{icst}^{g-1} \alpha_6 + \mathbf{X}_i \alpha_7 \\
& + \gamma_{cst}^{g-1} + \phi_s^g + \zeta_h + u_{icst}^h
\end{aligned} \tag{10}$$

This specification extends [Equation \(9\)](#) by adding the interaction of value-added measures with student gender. If the gender-differentiated effects documented in [Section 5.2](#) reflect teachers' heterogeneous impacts on students based on relative skill differences, the inclusion of Score RP $\times$ value-added should attenuate the Female $\times$ value-added coefficients. Specifically, comparing α_5 from [Equation \(10\)](#) to the Female $\times$ value-added coefficient from the unidimensional specifications reported in [Table 5](#) (Col. III in Panel A, and Col. VII in Panel B) provides an indication of the extent to which the gender-differentiated effect is captured by relative skill differences.

Table 6: Test of the Relative Skills Interpretation:
Seventh-Eighth Grade Outcomes

	Math Scores			Reading Grades		
	I	II	III	IV	V	VI
Female	0.218*** (0.080)	0.056*** (0.003)	0.217*** (0.080)	0.319*** (0.108)	0.301*** (0.004)	0.306*** (0.108)
TVA	0.089*** (0.029)	0.129*** (0.021)	0.091*** (0.029)	0.235*** (0.075)	0.135*** (0.050)	0.225*** (0.074)
Score RP		-0.001 (0.004)	-0.001 (0.004)		-0.135*** (0.006)	-0.134*** (0.006)
Female $\times$ TVA	0.084** (0.040)		0.074* (0.040)	-0.194* (0.100)		-0.180* (0.100)
Score RP $\times$ TVA		-0.093*** (0.035)	-0.087** (0.035)		0.125 (0.086)	0.116 (0.086)
Constant	-0.073 (0.205)	-0.071 (0.206)	-0.070 (0.204)	0.336 (0.249)	0.404 (0.248)	0.365 (0.248)
Mean	0.098	0.098	0.098	0.070	0.070	0.070
N	222,950	222,950	222,950	222,950	222,950	222,950
R ²	0.699	0.699	0.699	0.381	0.383	0.384

Note: Standard errors clustered at the fifth-grade teacher level. TVA refers to a 5th grade teacher's math score value-added (defined on 5th grade math scores) for columns I-III, and a 5th grade teacher's reading grade value-added (defined on 6th grade reading grades) for columns IV-VI. All outcomes are standardized (z-scores). Models include fourth-grade classroom fixed effects and fifth-grade school fixed effects. Stars denote significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Columns II and V of [Table 6](#) report the skill-heterogeneity test. I report results for middle school math scores and reading grades, which are the two outcomes with the strongest gender-differentiated effects in [Section 5.2](#) (results for all four outcomes are reported in [Appendix Table F.2](#)). For math scores, the interaction between test score

value-added and Score RP is negative and statistically significant at the one percent level (column II): teachers with who improve fifth-grade math test scores by 1 standard deviation have larger effects on the math test scores for students who have a relative deficiency in that dimension. For reading grades, the interaction between course grade value-added and Score RP is positive and larger in magnitude, but imprecisely estimated (column V). This pattern is consistent with (a) teachers who improve grades having stronger impacts on the students who have a relative deficiency in grades (i.e., students with high values of Score RP), and (b) course grades being noisier measures of student achievement than test scores.

The second test compares columns I and III for math scores and columns IV and VI for reading grades. For math scores, the Female $\times$ value-added coefficient shrinks from 0.084 standard deviations (column I) to 0.074 standard deviations (column III), a reduction of 0.01 standard deviations. For reading grades, it shrinks from -0.194 standard deviations (column IV) to -0.180 standard deviations (column VI), a reduction of 0.014 standard deviations. The movements are in the direction predicted by the framework but modest in magnitude. The modest magnitude is consistent with measurement error in Score RP, which is a discrete and therefore noisy proxy for relative skill differences.

Taken together, the results provide some support for the relative skills framework. The skill-heterogeneity test yields a significant interaction for math scores in the predicted direction; the reading-grade interaction has the predicted sign and a larger magnitude, but is imprecisely estimated. The second test shows modest attenuation of the gender-differentiated effects in both subjects, which is consistent with relative skill differences accounting for part of these effects. However, measurement error in Score RP suggests that the observed attenuation might be a lower bound.

5.4 Impacts on High School Outcomes

Elementary school teachers' effects on both test scores and grades predict long-term student educational attainment (Chetty, Friedman, and Rockoff, 2014b; Petek and Pope, 2023). I extend my analysis to high school outcomes, examining whether fifth-grade teachers' impacts on high school performance and graduation differ between boys and girls. A first outcome is whether students remain in the North Carolina public school system through high school. Graduation is observed only for students who appear in the state's cohort graduation rate (CGR) denominator, which is formed from first-time ninth-grade enrollment in North Carolina public schools (North Carolina Department of Public Instruction, 2025), and approximately 91 percent of my elementary and middle school sample meets this condition. Remaining in the public system is of interest in its own right; it is a precondition for the attainment outcomes I examine – a student must stay enrolled to graduate, enroll in college, or complete any later stage of schooling – and remaining in the school system reflects the non-cognitive skills like engagement and persistence, that shape educational attainment more broadly (Heckman, Stixrud, and Urzua, 2006). Furthermore, it also bears on

the analysis of graduation: if fifth-grade teachers affect whether students appear in the CGR sample – by affecting the likelihood of exiting the system before high school, or of otherwise not having a documented ninth-grade enrollment record – then sample membership is itself an outcome of the treatment, and estimates of graduation conditional on it may be contaminated by selection.

I first estimate [Equation \(7\)](#) on whether a student appears in the CGR sample. Column I of [Table 7](#) reports these estimates. A one-unit increase in fifth-grade teachers' course grade value-added increases the likelihood that boys remain in the system by 4.7 percentage points, while the effect for girls is close to zero; test score value-added predicts neither. The estimate does not clear conventional significance thresholds, so I read it as suggestive rather than definitive. However, the point estimate for boys is comparable to the effects on graduation reported below, and its standard error admits values large enough to matter for any analysis of graduation that conditions on sample membership.

I address this problem by estimating impacts on jointly-defined outcomes. Specifically, I construct an indicator that takes the value one if a student appears in the CGR sample and graduates, and zero otherwise. I define this joint outcome on two graduation windows – within four years and within six years of entering ninth-grade. Students who never appear in the CGR sample are coded as zero rather than dropped from the analysis, so [Equation \(7\)](#) is estimated on the full elementary and middle school sample, under the same assumptions as the analyses in [Section 5.2](#). This sidesteps the selection problem, since the sample is not conditioned on a variable that fifth-grade teachers may affect. The joint outcome is also the one most relevant for policy, since it is the unconditional probability that a student remains in the school system, and is recorded as a graduate. The coefficient on jointly-defined graduation reflects two channels, teachers' impacts on whether students are counted in the CGR sample, *and* their impacts on whether students graduate once counted. Comparing these coefficients to the estimates in Column I can give us a rough sense of the relative contribution of each channel.²⁵

Columns II and III of [Table 7](#) report impacts on jointly-defined graduation. A one-unit increase in fifth-grade teachers' course grade value-added increases the probability that boys are recorded as graduating within six years by 9.2 percentage points. The corresponding effect for girls is 2.2 percentage points, though the difference between the two is not precisely estimated. The effects on graduating within four years do not clear conventional significance thresholds; the four-year estimates are smaller, but directionally similar to the six-year estimates.²⁶ Since these outcomes combine staying in the school system with graduating, the estimated effect reflects both margins. A decomposition of the boys' six-year effect (reported in [Section F.3](#)) attributes roughly half to teachers' impact on

²⁵ [Appendix F.3.1](#) reports estimates conditional on CGR sample membership alongside the joint estimates, and offers a discussion on the relationship between the jointly-defined outcome and the conditional outcome.

²⁶ The observation count of 111317 students reflects the full analysis sample, less 158 singleton observations which are mechanically dropped by fixed-effects estimation.

Table 7: Gender-Differentiated Impacts of Fifth-Grade Teachers on High School Graduation

	In CGR Sample	Graduated (Joint)	
		Within 4 Years	Within 6 Years
	I	II	III
Female	0.009 (0.048)	0.075 (0.063)	0.083 (0.059)
Test Score VA	0.005 (0.026)	0.053 (0.034)	0.037 (0.032)
Female × Test Score VA	-0.027 (0.036)	-0.063 (0.048)	-0.033 (0.045)
Course Grade VA	0.047 (0.034)	0.066 (0.046)	0.092** (0.043)
Female × Course Grade VA	-0.041 (0.047)	-0.036 (0.063)	-0.070 (0.058)
Constant	1.069*** (0.133)	1.251*** (0.165)	1.169*** (0.159)
Mean	0.909	0.820	0.847
N	111317	111317	111317
r ²	0.114	0.152	0.151

Note: Standard errors clustered at the fifth-grade teacher level. Column I reports impacts on an indicator for appearing in North Carolina's cohort graduation rate (CGR) sample. Columns II and III report impacts on jointly-defined outcomes, each taking the value one if a student appears in the CGR sample *and* graduates within the stated window, and zero otherwise. Test Score VA is defined on fifth-grade test scores. Course Grade VA is defined on sixth-grade course grades. Models include fourth-grade classroom fixed effects and fifth-grade school fixed effects. Stars denote significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

graduation conditional on remaining in the public school system, and roughly half to their impact on remaining in the system itself, though neither component is precisely estimated on its own.

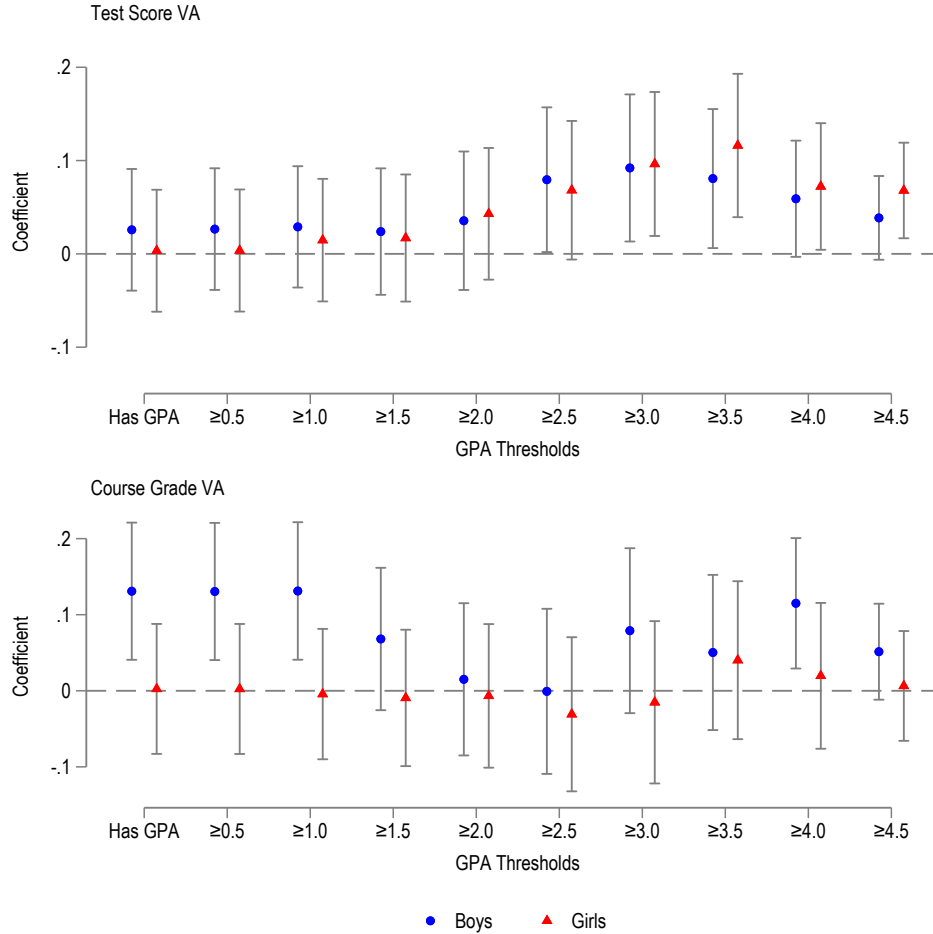
The sample restriction applies to high school GPA as well: a student's weighted GPA is recorded only if they appear in the CGR sample and complete the required coursework to generate a transcript. I therefore apply the same approach of jointly defining GPAs with students' presence in the CGR sample. I construct a series of threshold indicators at half-point intervals across the six-point GPA scale, each taking the value one if a student appears in the CGR sample *and* has a weighted GPA at or above a given cutoff, and zero otherwise. This preserves the joint-outcome logic while allowing teachers' impacts to differ across the GPA distribution. I estimate [Equation \(7\)](#) for each threshold indicator, as well as for an indicator for whether the student had a recorded GPA.

[Figure 5](#) plots these estimates for boys and girls, separately for test score and course grade value-added measures. Fifth-grade teachers' course grade value-added increases the probability that boys have a recorded high school GPA by 13.1 percentage points, an effect that is close to zero for girls.²⁷ This margin is essentially the same as appearing in the CGR sample with a transcript: almost no student with a recorded GPA has one below 1.0, so the lowest thresholds are nearly indistinguishable from the indicator for having a GPA at all. Above that margin, the pattern

²⁷ [Appendix F.3.2](#) reports the underlying coefficients.

reverses. Test score value-added predicts increases through the middle and upper portions of the GPA distribution, raising the probability of a GPA at or above 2.5, 3.0, and 3.5 by between 7.9 and 9.2 percentage points, with similar effects for boys and girls.

Figure 5: Gender-Differentiated Impacts of Fifth-Grade Teachers Across the High School GPA Distribution



Note. Each panel plots coefficients from estimating Equation (7) on jointly-defined GPA threshold indicators, each taking the value one if a student appears in the CGR sample and has a weighted GPA at or above the indicated cutoff, and zero otherwise. “Has GPA” denotes an indicator for appearing in the CGR sample with any recorded GPA. The top panel reports impacts of test score value-added; the bottom panel reports impacts of course grade value-added. Blue circles report estimates for boys and red triangles report estimates for girls; bars denote 95 percent confidence intervals. Standard errors are clustered at the fifth-grade teacher level. Underlying coefficients are reported in Appendix F.3.2.

Taken together, the results suggest that test score value-added is the better predictor of GPA at the upper tail of the distribution, while course grade value-added is what predicts whether students have a recorded GPA at all. This mirrors what Backes et al. (2024) find at the postsecondary level, where test-based measures of teacher quality better explain students enrolling at selective colleges, while non-test measures better explain whether students go to any college at all. My estimates further suggest that these margins are gendered: the effects on having a GPA recorded are concentrated among boys, while the upper-tail effects are approximately equal for boys and girls. This is consistent with the effects on graduation – since students who are in the upper-tail of the GPA distribution aren’t

typically on the margin for graduating.

6 Conclusion

I document that teachers' impacts on student outcomes are multidimensional and systematically vary by gender. Teachers with high value-added in test scores improve girls' outcomes more than they improve boys' outcomes (both test scores and grades), while teachers with high value-added in course grades improve boys' outcomes more than they improve girls' outcomes. I interpret these patterns through a framework of relative skill differences, in which test scores are relatively more intensive in cognitive skills, course grades are relatively more intensive in non-cognitive skills, and teachers who are effective at improving a given skill have stronger impacts on the students who have a relative deficiency in that skill. This interpretation of test scores and grades is supported by several strands of prior evidence. Gender gaps in grades persist (and expand) even after conditioning on test scores; they predict gender gaps in long-run outcomes like high school graduation and college attendance; and they are explained by gender gaps in other non-cognitive skills like approaches to learning (Cornwell, Mustard, and Van Parys, 2013; Jacob, 2002). Under this framework, observed gender gaps in test scores and course grades reflect gender gaps in the ratio of latent skills. Specifically, boys have a relative proficiency in cognitive skills, and girls have a relative proficiency in non-cognitive skills. As a result, teachers who improve test scores have stronger impacts on girls, and teachers who improve course grades have stronger impacts on boys. This interpretation differs from prior work in the teacher value-added literature that emphasizes role-model effects or biased grading (Barrios-Fernández and Riudavets-Barcons, 2024; García-Echalar, Poblete, and Rau, 2024). The empirical results are directionally consistent with framework's predictions, both for the gender-differentiated impacts documented in Section 5.2 and for the individual-level test using students' relative skill profiles in Section 5.3). Attenuation in the estimates reported in Section 5.3 is consistent with measurement error in the relative-skills measure, which is necessarily discrete, suggesting that the documented patterns understate a likely stronger underlying relationship.

These findings have implications for how we measure and interpret teacher quality. Evaluating teachers along a single metric (typically test score value-added) misses dimensions of teacher effectiveness that matter for students' long-run outcomes. I show that teachers who are effective at improving course grades not only have stronger impacts on boys' middle school outcomes than girls' outcomes, but also raise the likelihood of boys remaining in the public school system and graduating high school within six years, with a smaller effect for girls (Section 5.4). Since boys benefit more from course grade value-added (Section 5.2), evaluating teachers on just their value-added in test scores may have distributional consequences; the teachers whose effectiveness benefits boys' educational achievement the most are not the teachers identified by test score value-added alone. While prior literature does speak to the importance of evaluating teachers based on their impacts on non-cognitive skills (Jackson, 2018; Petek and Pope,

2023; Rose, Schellenberg, and Shem-Tov, 2022), I show that among the typically used non-cognitive measures, course grade value-added is the only one that passes both validation and placebo tests in my sample. Behavioral measures (absences, suspensions, and grade repetition) either fail standard validation checks or lack sufficient variation to be reliably estimated (Section 3.3). As a result, this paper suggests that course grade value-added may be used as a standalone measure of teacher effectiveness, rather than absorbed into a composite behavioral index, as is common in the literature.

An underlying caveat in the findings is that of measurement error, which likely understates the strength of my findings. The relative-skills measure used in the direct test of the framework (Section 5.3) is necessarily discrete, since course grades take on discrete values, limiting the precision with which Score RP can capture students' underlying relative skill profiles. Course grades themselves are anticipated rather than finalized in the data, introducing measurement error into both the dependent and independent variables in several specifications. As a result, several gender-differentiated patterns (specifically in the subject-specific models) are also imprecisely estimated, in part because the relevant subsamples are smaller and noisier. Across these constraints, the attenuation and imprecision in the estimates is consistent with measurement error in both the outcomes and the explanatory variables, suggesting the true underlying relationships may be stronger than the point estimates imply.

Several avenues for future research remain open. First, while the specific classroom practices that generate the underlying variation in teacher effectiveness are typically not observed in administrative data, this paper's findings shed light on the structure of those practices. Teachers' impacts vary with students' underlying skill profiles, which correlate with but are not reducible to student gender. Future work examining settings where teacher evaluation incorporates classroom observation or multidimensional metrics could identify the specific practices that could result in heterogeneous impacts by students' underlying skill mixes. Second, the gender-differentiated effects documented in this paper have implications for student-teacher assignment. Delgado (2025) and Eastmond et al. (2026) show that allocating teachers based on heterogeneous value-added measures (by race, gender, or prior achievement) can generate substantial welfare gains over the status quo. The findings in this paper suggest a unifying interpretation: demographic markers and prior achievement work as dimensions for heterogeneity in teacher effects, since *both* proxy for students' underlying relative skill profiles. Direct measurement of these underlying skill profiles may further refine how we think about optimal assignment. Third, the long-run findings in Section 5.4 extend the gender-differentiated patterns from middle school to high school, specifically on high-school GPA and graduation. Testing whether the patterns extend to outcomes such as college attendance, completion, and labor market outcomes is a natural direction for future work. Finally, future research could explore whether these patterns extend to other settings, particularly in contexts where grading conventions or gender norms differ. The relative skills framework predicts the direction of gender-differentiated teacher impacts based on the structure of gender gaps in test scores

and course grades; settings with different baseline gaps could provide independent tests of the framework.

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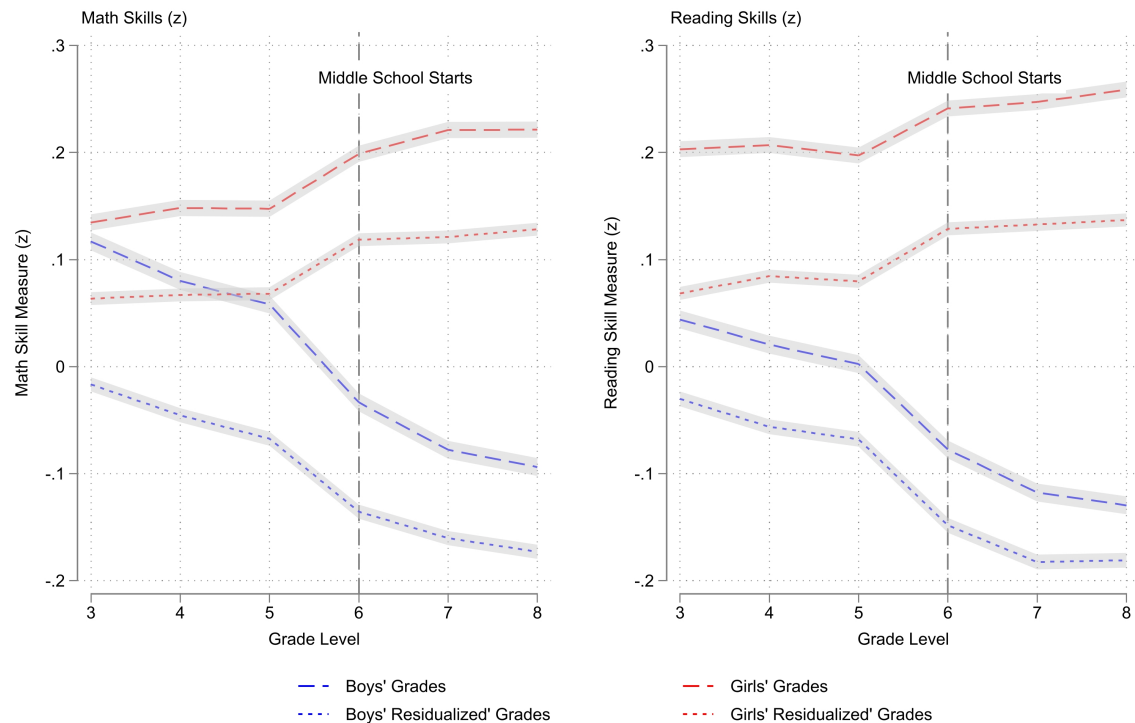
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A Progression of Gender Gaps in Course Grades

Figure A.1 plots boys' and girls' course grades in math and reading from third through eighth grade, alongside residualized grades obtained by regressing course grades on test scores within each subject. Gender gaps in grades persist (and expand) even after conditioning on test scores (dotted lines), suggesting that grades capture a set of skills that are not adequately represented by test scores – skills in which boys on average lag behind girls.

Figure A.1: Math and Reading Skills: Evolution of Gender Gaps
Third–Eighth Grade



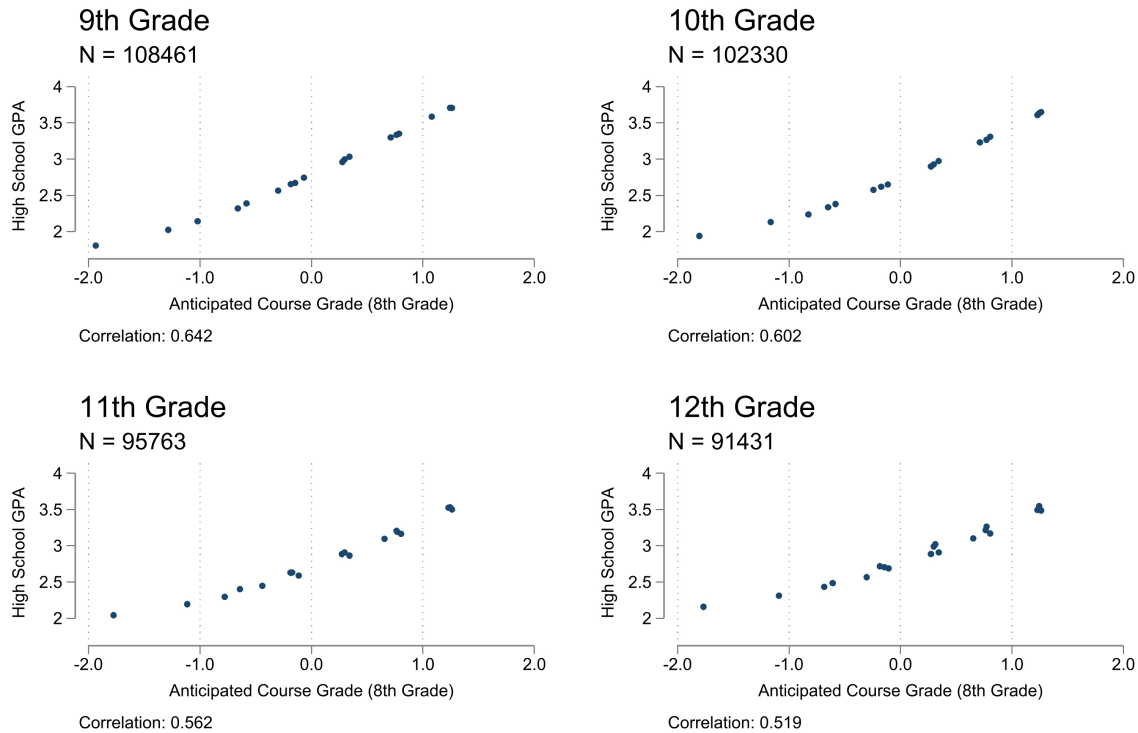
Note. Test scores and course grades are standardized within grade-year. The sample is restricted to students who can be traced from third-eighth grade. Residualized grades are the residuals extracted from regressing course grades on test scores within each subject. I describe more details about the sample restrictions in [Section 2.1](#).

B Summary Statistics and Descriptive Relationships

B.1 Anticipated Grades and High School GPAs

Figure B.1 plots high school GPA against eighth-grade anticipated course grades (averaged across math and reading), separately for ninth through twelfth grade. Anticipated grades in 8th grade are strongly correlated with actual GPA's in high school, with correlation coefficients ranging from 0.51 to 0.65.

Figure B.1: High School GPAs vs Eighth Grade Anticipated Course Grades
Ninth-Twelfth Grade



Note. Unweighted high school GPA's for each grade in high school plotted against the average (anticipated) grade for math and reading from eighth grade. Points represent binned means. Correlations are calculated at the student level for all students with an observed unweighted GPA in the respective grade.

B.2 Sample Selection and Summary Statistics for Analysis Sample

Table B.1 reports summary statistics for the unrestricted sample of traceable students, alongside differences between students included in and excluded from the analysis sample. Students in the analysis sample are positively selected relative to the full set of traceable students: at the end of fourth grade, they have higher test scores and course grades, and a stronger behavioral profile. The analysis sample also contains fewer economically disadvantaged, ESL, and disabled students, fewer Black, Hispanic, and Asian students, and somewhat larger classes, though these demographic differences are modest.

Table B.1: Summary Statistics: Analysis Sample vs. Unrestricted Sample (Fourth Grade)

Student Outcomes			Student Characteristics		
	Mean	Diff		Mean	Diff
Test Score (z)	0.009 (0.929)	0.164*** (0.003)	Female	0.494 (0.500)	0.031*** (0.002)
Math (z)	0.010 (0.995)	0.179*** (0.004)	White	0.549 (0.498)	0.055*** (0.002)
Reading (z)	0.002 (0.996)	0.158*** (0.004)	Black	0.266 (0.442)	-0.036*** (0.002)
Course Grade (z)	0.009 (0.940)	0.171*** (0.003)	Hispanic	0.106 (0.308)	-0.015*** (0.001)
Math (z)	0.009 (0.996)	0.171*** (0.004)	Asian	0.022 (0.148)	-0.003*** (0.001)
Reading (z)	0.008 (0.995)	0.174*** (0.004)	Other	0.057 (0.231)	-0.001 (0.001)
Behavioral Skills Index (z)	-0.003 (0.999)	0.142*** (0.004)	Disadvantaged	0.492 (0.500)	-0.045*** (0.002)
ln(1+Absences)	1.672 (0.825)	-0.083*** (0.003)	ESL	0.067 (0.250)	-0.018*** (0.001)
Suspended	0.046 (0.209)	-0.018*** (0.001)	Reported Disability	0.139 (0.346)	-0.073*** (0.001)
Repeated Grade	0.010 (0.101)	-0.009*** (0.000)	Class Size	21.559 (4.847)	0.406*** (0.017)

Note: Reported means are for the unrestricted sample of students who can be traced from third through eighth grade. Standard deviations are reported in parentheses below the means. Differences are computed as the mean for students in the analysis sample minus the mean for traceable students not included in the analysis sample, with standard errors of the difference in parentheses below. All test scores and grades are standardized within grade-year (z-scores). The Behavioral Skills Index is a standardized composite combining absences, suspensions, and grade repetition, following the methodology of Jackson (2018). Stars denote significance levels for a t-test for differences in means: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

B.3 Behavioral Outcomes

Table B.2 reports gender differences in behavioral outcomes for the analysis sample, separately for elementary (third-fifth grade) and middle school (sixth-eighth grade).

B.4 Summary Statistics: Teacher Characteristics

Table B.3 compares teacher characteristics in the analysis sample to the full set of North Carolina elementary school teachers. Analysis-sample teachers are more experienced than the full set; the analysis sample includes teachers for whom leave-cohort-out value-added measures were constructed, which restricts attention to teachers who taught for a minimum of two years in the North Carolina public school system.

Table B.2: Summary Statistics: Behavioral Outcomes (Analysis Sample)

	Elementary School (3rd-5th Grade)		Middle School (6th-8th Grade)	
	Mean	Diff	Mean	Diff
Behavioral Skills Index (z)	0.086 (0.893)	-0.115*** (0.003)	0.080 (0.911)	-0.184*** (0.003)
ln(1+Absences)	1.626 (0.812)	0.012*** (0.003)	1.709 (0.852)	0.033*** (0.003)
Suspended	0.034 (0.180)	0.038*** (0.001)	0.100 (0.300)	0.073*** (0.001)
Repeated Grade	0.007 (0.082)	0.001*** (0.000)	0.004 (0.060)	0.003*** (0.000)
N = 111,475				

Note: Reported means are for the analysis sample, with standard deviations in parentheses below. Differences are computed as the boys' mean minus the girls' mean in the analysis sample, with standard errors of the difference in parentheses below. N denotes the number of students in the analysis sample; summary statistics are computed at the student-grade level, pooled within each schooling level. The Behavioral Skills Index is a standardized composite combining absences, suspensions, and grade repetition, following Jackson (2018). Stars denote significance levels for a t-test for differences in means: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B.3: Summary Statistics: Teacher Characteristics

	Full Set	Analysis Sample
Female	0.88 (0.33)	0.87 (0.34)
Black	0.14 (0.35)	0.14 (0.34)
Hispanic	0.01 (0.09)	0.01 (0.08)
White	0.85 (0.36)	0.85 (0.35)
Experience (Years)	9.08 (5.58)	10.08 (5.33)
Graduate Degree	0.34 (0.48)	0.35 (0.48)
Observations	8,591	4,920

Note: Standard deviations in parentheses. The analysis sample is restricted based on the conditions described in [Section 2.1](#).

C Summary Statistics and Descriptive Relationships for Value-Added Measures

C.1 Behavioral Skills Value-Added Measures

I construct value-added measures for three behavioral outcomes (absences, suspensions, and grade repetition), following the procedure in [Section 3.1](#). Using the lead outcome identifies teacher value-added from the persistence of effects into the subsequent grade and reduces bias from teachers grading or evaluating their own students, following Jackson (2018) and Petek and Pope (2023). The vector of prior outcomes $\mathbf{Y}_{icst}^{g-1}$ includes test scores and course grades averaged across subjects, together with lagged behavioral measures. The estimation procedure follows Jackson (2018) and Petek and Pope (2023). I also construct value-added for two composite behavioral indices. The four-component index (Behavioral Skills + Grades) combines course grades with absences, suspensions, and grade repetition, following Jackson (2018) directly. The three-component index (Behavioral Skills) combines only the three behavioral measures. Both indices are standardized, and value-added is estimated for each using the same procedure as for the individual measures.

C.2 Summary Statistics: Behavioral Value-Added Measures

[Table C.1](#) reports standard deviations and correlations of the behavioral value-added measures. The behavioral measures are more dispersed than the academic measures: the standard deviations of value-added for absences, suspensions, and grade repetition (0.160, 0.181, and 0.192) all exceed that of test score value-added (0.104), and the two composite indices are similarly dispersed. The components correlate weakly with one another and with course grade value-added: the largest pairwise correlation among them is 0.130, and the three-component index correlates with course grade value-added at 0.183. Behavioral value-added thus captures variation that is largely distinct from the academic measures, consistent with Petek and Pope (2023).

Table C.1: Standard Deviations and Correlations of Behavioral Value-Added Measures

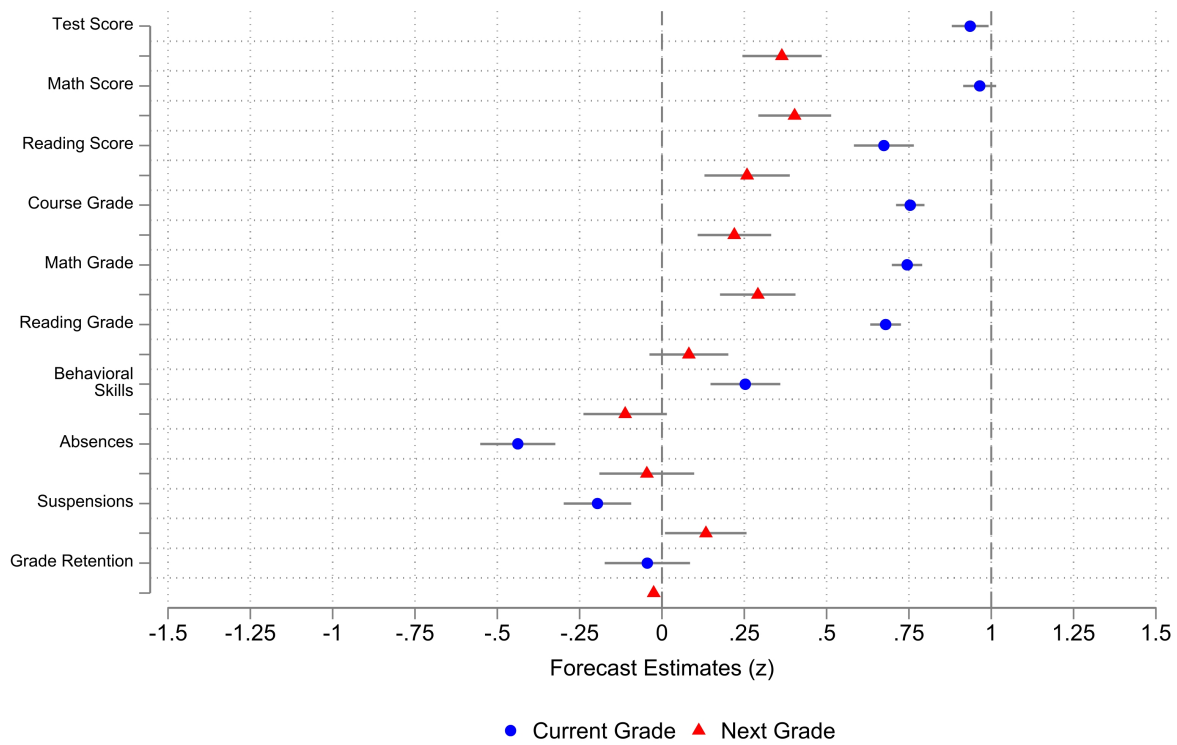
<i>Panel A: Standard Deviations</i>				
	SD			
Behavioral Skills + Grades (z)	0.159			
Course Grade (z)	0.132			
Behavioral Skills (z)	0.172			
Absences (z)	0.160			
Suspensions (z)	0.181			
Grade Repetition (z)	0.192			
<i>Panel B: Correlations, Component Measures</i>				
	Course Grade	Absences	Suspensions	Grade Repetition
Course Grade (z)	1			
Absences (z)	0.130	1		
Suspensions (z)	0.115	0.127	1	
Grade Repetition (z)	0.064	0.030	0.077	1
Correlation between Course Grade VA and Behavioral Skills VA (three-component): 0.183				

Note: Panel A reports standard deviations of fifth-grade teacher value-added measures, expressed in standard deviations of student outcomes. Standard deviations are reported for the unshrunk value-added measures, to facilitate comparison with the literature. The four-component index combines course grades with the three-component behavioral skills index; the three-component index combines absences, suspensions, and grade repetition, following the methodology of Jackson (2018). Panel B reports correlations among the component measures. All value-added measures are estimated as described in [Section 3.1](#).

C.3 Validity Tests: All Measures

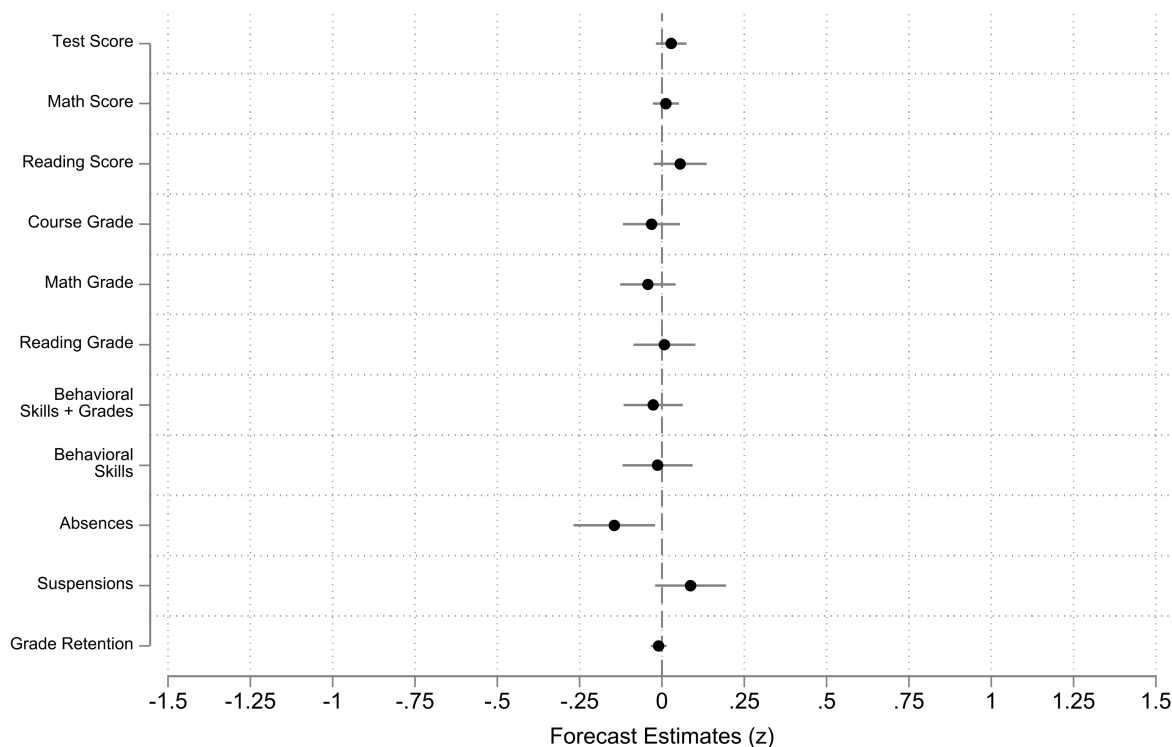
[Figure C.1](#) reports forecast-unbiasedness estimates for all value-added measures, and [Figure C.2](#) reports the corresponding placebo-test estimates. The forecast-unbiasedness pattern matches that for the focal measures discussed in [Section 3.3](#): contemporaneous estimates are largest for test score value-added and attenuated for the lead-based measures, and the attenuation is general across the subject-specific academic measures and the behavioral measures alike. On the placebo test, test score value-added and course grade value-added do not predict prior-grade outcomes, and the same holds for the subject-specific academic measures and the two composite indices. Among the individual behavioral components, value-added for absences significantly predicts prior-grade absences, and value-added for suspensions predicts prior-grade suspensions at the ten percent level. These components therefore fail the placebo test, indicating that they partly reflect student sorting rather than teacher effectiveness. While value-added for grade repetition does not predict prior retention, grade repetition is too rare to yield reliable estimates, as discussed in [Section 3.3](#).

Figure C.1: Forecast Bias in Value-Added Measures
Fifth-Grade Teachers



Note. Figure reports estimated values of $\hat{\alpha}_1$ for value-added measures constructed for test scores, course grades, and behavioral outcomes, measured in standard deviation units of the outcomes. Estimated coefficients are from regressions wherein 5th (current) and 6th (next) grade outcomes are regressed on 5th grade VA measures.

Figure C.2: Placebo Test for Sorting on Prior Gains
Fifth-Grade Teachers

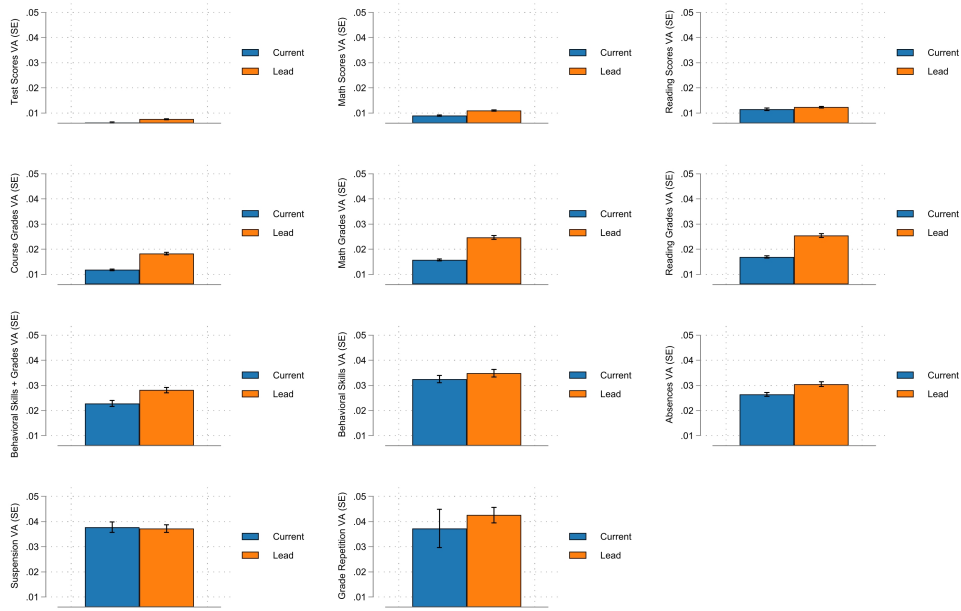


Note. Figure reports estimated values of $\hat{\alpha}_1$ for value-added measures constructed for test scores, course grades, and behavioral outcomes, measured in standard deviation units of the outcomes. Estimated coefficients are from regressions wherein 4th (prior) grade outcomes are regressed on 5th grade VA measures.

C.4 Standard Errors of Unshrunk Fixed Effects

Figure C.3 plots the standard errors of the unshrunk value-added measures, separately for contemporaneous ($h = g$) and lead ($h = g + 1$) outcomes. Two patterns are evident. First, standard errors are larger for the lead measures than for the contemporaneous measures across the academic measures and the composite behavioral indices; the two exceptions are suspensions and grade repetition, whose value-added is imprecisely estimated (with very large standard errors throughout). Second, standard errors are smallest for test score value-added, larger for course grade value-added, and largest for the behavioral measures. Both patterns are consistent with the attenuation of the forecast-unbiasedness estimates discussed in Section 3.3: lead-based and non-test score measures are noisier, and noisier measures are more attenuated and more heavily shrunk.

Figure C.3: Standard Errors of Unshrunk Teacher Fixed Effects
Fifth-Grade Teachers



Note. Figure plots standard errors of unshrunk fixed effects for all value-added measures, constructed for contemporaneous ($h = g$, blue bars) and lead ($h = g + 1$, orange bars) outcomes, for 5th grade teachers. Higher means of standard errors indicate noisier estimates, and therefore greater shrinkage towards the mean.

C.5 Behavioral Skills Value-Added Regressions

Table C.2 regresses middle-school behavioral outcomes on fifth-grade teachers' value-added measures. Each block adds value-added measures progressively: the matching behavioral component's own value-added, then course grade value-added, then test score value-added. Across all specifications, value-added for each behavioral component fails to predict that component's own outcome later in middle school: the coefficients on absences, suspensions, and grade repetition value-added are small and statistically insignificant throughout. A teacher identified as reducing a given behavioral outcome does not measurably reduce that outcome in subsequent grades.

Test score value-added predicts two of the three behavioral outcomes; it is associated with fewer subsequent absences and, in the opposite direction, more subsequent suspensions. On the other hand, course grade value-added is largely unrelated to them. The inconsistent signs for the effects of teachers' test score value-added do not support treating the measure as a proxy for teachers' effects on behavior.

Table C.2: Multidimensional Impacts of Fifth-Grade Teachers on Middle School Behavioral Outcomes

	Absences (6th-7th Grade)			Suspended (6th-8th Grade)			Repeated Grade (6th-8th Grade)		
	I	II	III	IV	V	VI	VII	VIII	IX
Test Score VA			-0.112*** (0.042)			0.077** (0.038)			0.012 (0.025)
Course Grade VA		-0.037 (0.057)	-0.006 (0.058)		-0.047 (0.051)	-0.069 (0.052)		0.057* (0.034)	0.053 (0.035)
Behavioral VA	0.008 (0.059)	0.013 (0.059)	0.018 (0.059)	-0.036 (0.040)	-0.032 (0.041)	-0.029 (0.041)	-0.039 (0.031)	-0.044 (0.031)	-0.043 (0.031)
Constant	-0.382 (0.268)	-0.381 (0.268)	-0.384 (0.268)	0.114 (0.357)	0.114 (0.357)	0.115 (0.357)	0.340* (0.195)	0.339* (0.195)	0.339* (0.195)
Mean	-0.048	-0.048	-0.048	-0.068	-0.068	-0.068	-0.039	-0.039	-0.039
N	222950	222950	222950	334425	334425	334425	334425	334425	334425
R ²	0.321	0.321	0.321	0.171	0.171	0.171	0.046	0.046	0.046

Note: Standard errors clustered at the fifth-grade teacher level. Test Score VA defined on fifth-grade test scores. Course grade and Behavioral VA measures defined on sixth-grade outcomes. Behavioral VA is Absences VA for columns I-III, Suspensions VA for columns IV-VI, and Grade Repetition VA for columns VII-IX. All outcomes are z-scores. All models include fourth-grade classroom fixed effects and fifth-grade school fixed effects. Stars denote significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

D Gender-Differentiated Impacts

Teachers can produce improvements in students' outcomes through their independent effects on cognitive skills (c) or non-cognitive skills (n). These effects are expressed by the following partial derivatives of test scores (y) and grades (z):

$$\frac{\partial y_{\kappa s}}{\partial c_{\kappa s}} = \theta \left(\frac{1}{\psi_{\kappa s}} \right)^{1-\theta}, \quad (\text{D.1})$$

$$\frac{\partial z_{\kappa s}}{\partial c_{\kappa s}} = \gamma \left(\frac{1}{\psi_{\kappa s}} \right)^{1-\gamma}, \quad (\text{D.2})$$

$$\frac{\partial y_{\kappa s}}{\partial n_{\kappa s}} = (1 - \theta) \psi_{\kappa s}^{\theta}, \quad (\text{D.3})$$

$$\frac{\partial z_{\kappa s}}{\partial n_{\kappa s}} = (1 - \gamma) \psi_{\kappa s}^{\gamma}, \quad (\text{D.4})$$

where $\psi_{\kappa s} = c_{\kappa s}/n_{\kappa s}$ denotes the relative skill mix for gender $\kappa \in \{B, G\}$ in subject s .

D.1 Teachers Who Improve Cognitive Skills

1. On test scores:

$$\frac{\text{girls' improvement}}{\text{boys' improvement}} = \frac{\theta \left(\frac{1}{\psi_{Gs}} \right)^{1-\theta}}{\theta \left(\frac{1}{\psi_{Bs}} \right)^{1-\theta}} = \left(\frac{\psi_{Bs}}{\psi_{Gs}} \right)^{1-\theta} > 1. \quad (\text{D.5})$$

2. On grades:

$$\frac{\text{girls' improvement}}{\text{boys' improvement}} = \frac{\gamma \left(\frac{1}{\psi_{Gs}} \right)^{1-\gamma}}{\gamma \left(\frac{1}{\psi_{Bs}} \right)^{1-\gamma}} = \left(\frac{\psi_{Bs}}{\psi_{Gs}} \right)^{1-\gamma} > 1. \quad (\text{D.6})$$

⇒ Girls improve more.

D.2 Teachers Who Improve Non-Cognitive Skills

1. On test scores:

$$\frac{\text{boys' improvement}}{\text{girls' improvement}} = \frac{(1-\theta) \psi_{Bs}^{\theta}}{(1-\theta) \psi_{Gs}^{\theta}} = \left(\frac{\psi_{Bs}}{\psi_{Gs}} \right)^{\theta} > 1. \quad (\text{D.7})$$

2. On grades:

$$\frac{\text{boys' improvement}}{\text{girls' improvement}} = \frac{(1-\gamma) \psi_{Bs}^{\gamma}}{(1-\gamma) \psi_{Gs}^{\gamma}} = \left(\frac{\psi_{Bs}}{\psi_{Gs}} \right)^{\gamma} > 1. \quad (\text{D.8})$$

⇒ Boys improve more.

E Gender-Specific Value-Added

This appendix presents the construction, descriptive properties, empirical specification, and results for the gender-specific value-added measures used as a supplementary check on the gender-differentiated findings in [Section 5.2.1](#). The analysis follows the approach of Barrios-Fernández and Riudavets-Barcons (2024), who construct teacher value-added measures separately for each gender of student and use them to test for gender-heterogeneous teacher effects. [Section E.1](#) describes how the measures are constructed. [Section E.2](#) reports two descriptive relationships between the boy-specific and girl-specific measures. [Section E.3](#) describes the empirical specification. [Section E.4](#) reports the results.

E.1 Construction of Gender-Specific Value-Added Measures

Boy-specific and girl-specific value-added measures for a fifth-grade teacher are constructed using the same estimation procedure as the pooled value-added measures described in [Section 3.1](#), but restricting the estimation sample to one gender of students at a time. To ensure sufficient support within both groups, I further restrict the estimation sample to classrooms containing a minimum of six boys and six girls. For each fifth-grade teacher j , the boy-specific value-added measure $\widehat{\delta}_j^{g,B}$ is estimated using only the boys taught by j in grade g , and the girl-specific measure $\widehat{\delta}_j^{g,G}$ is estimated using only the girls taught by j in grade g . Both measures are constructed for test scores (using fifth-grade test scores) and course grades (using sixth-grade course grades), yielding two pairs of measures per teacher: $\delta_j^{g,B} = [\widehat{\delta}_j^{g,B, \text{score}}, \widehat{\delta}_j^{g,B, \text{grade}}]$ for boys, and the corresponding pair $\delta_j^{g,G}$ for girls.

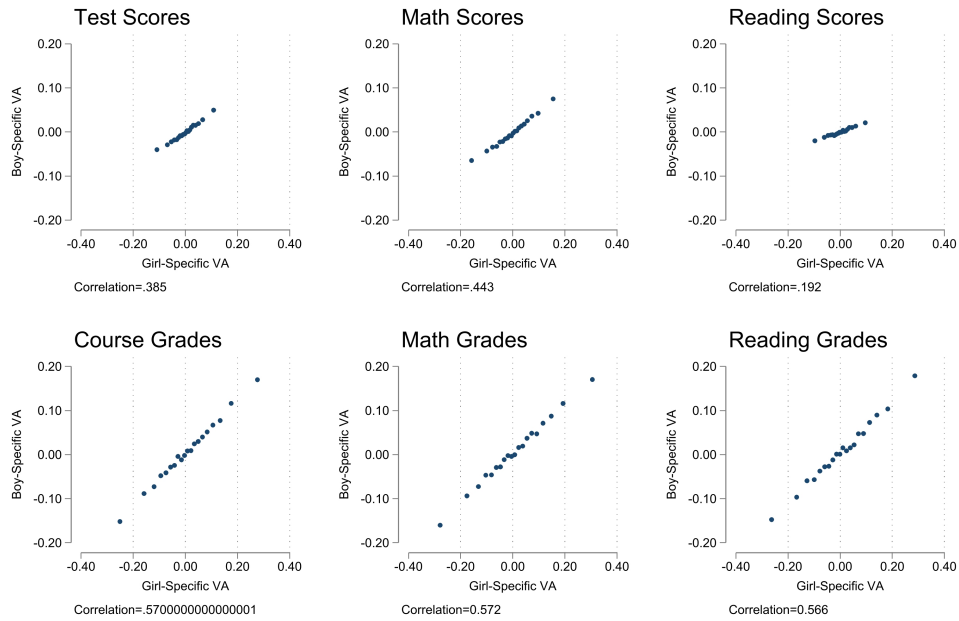
The estimation follows the same procedure as in [Section 3.1](#): leave-cohort-out unshrunk fixed effects, shrunk via the empirical Bayes procedure described therein, with the signal variances and empirical Bayes shrinkage factors estimated separately within each gender. Subject-specific gender-specific measures (boy-specific math score VA, girl-specific reading grade VA, and so on) are constructed analogously for the subject-level analyses.

E.2 Descriptive Relationships

[Figure E.1](#) plots boy-specific against girl-specific value-added across teachers. The two are only moderately correlated: the correlation is around 0.4 for test score value-added and roughly 0.57 for course grade value-added, with similar magnitudes for the subject-specific measures. A teacher's effectiveness for one gender is therefore an imperfect predictor of her effectiveness for the other, and more so for test scores than for course grades.

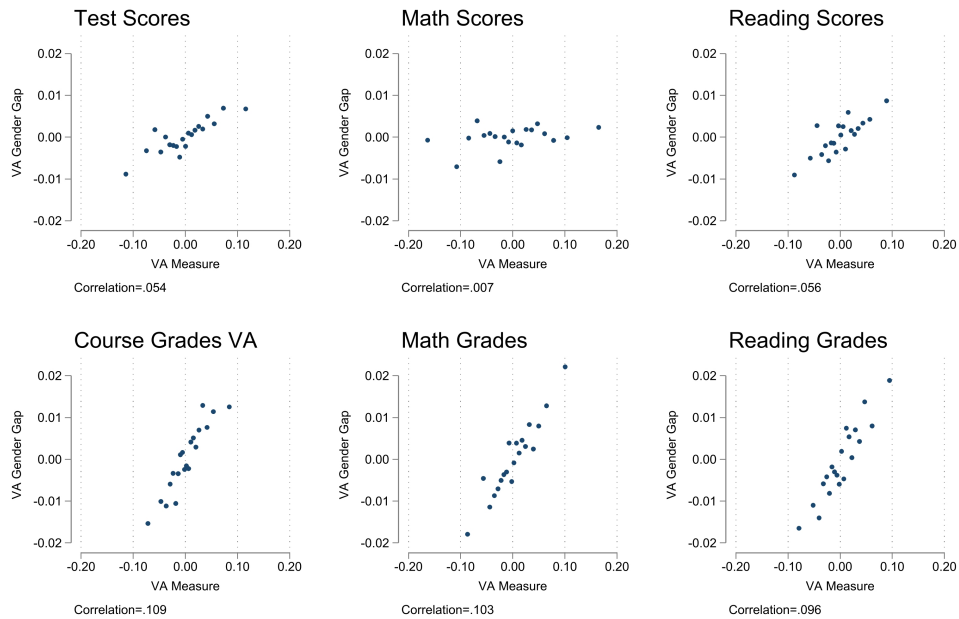
[Figure E.2](#) plots each teacher's gender gap in value-added (the difference between her boy-specific and girl-specific measures) against her overall value-added. The two are essentially uncorrelated for every measure. Highly effective teachers do not seem to systematically favor one gender over the other.

Figure E.1: Boy-Specific vs Girl-Specific Value-Added
Fifth-Grade Teachers



Note. Figure plots binned scatterplots of boy-specific value-added measures against girl-specific value-added measures of 5th grade teachers for test scores and course grades. Boy (girl)-specific value-added is defined as a teacher value-added measure based on only the boys (girls) taught by a 5th grade teacher, as described in [Section E.1](#).

Figure E.2: Value-Added Gender Gaps
Fifth-Grade Teachers



Note. Figure plots binned scatterplots of value-added gender gaps against the overall value-added measures of 5th grade teachers for test scores and course grades. Value-added gender gaps are defined as the difference between a 5th grade teacher's boy-specific value-added measure and girl-specific value-added measure.

E.3 Empirical Specification

To test whether teachers' boy-specific and girl-specific value-added have heterogeneous impacts on boys' and girls' outcomes, I estimate the specification:

$$y_{icst}^h = \alpha_0 + (\delta_j^{g,B} \times M_i) \alpha_1 + (\delta_j^{g,G} \times M_i) \alpha_2 + (\delta_j^{g,B} \times F_i) \alpha_3 + (\delta_j^{g,G} \times F_i) \alpha_4 + \mathbf{Y}_{icst}^{g-1} \alpha_5 + \mathbf{X}_i \alpha_6 + \gamma_{cst}^{g-1} + \phi_s^g + u_{icst}^h \quad (\text{E.1})$$

Here F_i is an indicator for whether student i is female and $M_i = 1 - F_i$ is an indicator for whether student i is male. The vector $\delta_j^{g,B}$ denotes teacher j 's boy-specific value-added in grade g , constructed as in [Section E.1](#), and $\delta_j^{g,G}$ denotes the corresponding girl-specific value-added.

The controls $\mathbf{Y}_{icst}^{g-1}$ and $\mathbf{X}_i$ are defined as in [Section 3.1](#): $\mathbf{Y}_{icst}^{g-1}$ contains lagged subject-specific test scores, course grades, absences, suspensions, and grade-repetition flags for student i in grade $g - 1$, together with leave-one-out classroom and school-level averages of those outcomes, all entering as cubic polynomials; $\mathbf{X}_i$ contains student i 's race, ethnicity, English-as-a-second-language status, economic disadvantage, and disability status. As in the baseline specification, every control is interacted with the female indicator, so that the relationship between prior achievement, demographics, and current outcomes is allowed to differ for boys and girls.

The coefficient vectors $\alpha_1, \alpha_2, \alpha_3, \alpha_4$ each have a score and a grade component (e.g., $\alpha_1 = [\alpha_1^{\text{score}}, \alpha_1^{\text{grade}}]$) and capture the four student-teacher gender interactions: α_1 is the effect of boy-specific value-added on boys, α_2 the effect of girl-specific value-added on boys, α_3 the effect of boy-specific value-added on girls, and α_4 the effect of girl-specific value-added on girls. Comparing these coefficients tests whether the gender-differentiated impacts documented in [Section 5.2](#) reflect heterogeneity in teacher effectiveness by gender of student. If teachers who effectively raise test scores benefit girls more than boys (as predicted by the relative skills framework), then the total effect of test score value-added on girls should exceed the total effect on boys: $\alpha_3^{\text{score}} + \alpha_4^{\text{score}} > \alpha_1^{\text{score}} + \alpha_2^{\text{score}}$. Symmetrically, the framework predicts $\alpha_1^{\text{grade}} + \alpha_2^{\text{grade}} > \alpha_3^{\text{grade}} + \alpha_4^{\text{grade}}$ for course grade value-added. As in the baseline specification, these predicted inequalities hold whether the outcome y_{icst}^h is a test score or a course grade.

E.4 Results

[Table E.1](#) reports estimates of [Equation \(E.1\)](#) pooled across subjects, and [Table E.2](#) reports the corresponding estimates separately for math (Panel A) and reading (Panel B). The patterns are consistent across specifications. For test score value-added, the boy-specific measure predicts test score improvements for both genders, while the girl-specific measure predicts improvements primarily for girls. For course grade value-added, the pattern reverses:

the girl-specific measure predicts course grade improvements for both genders, while the boy-specific measure predicts improvements primarily for boys. These asymmetric cross-impacts are observed in both math and reading.

Table E.1: Gender-Differentiated Impacts of Fifth-Grade Teachers on Middle School Outcomes

	Test Scores		Course Grades	
	I	II	III	IV
Male $\times$ Boy-Specific Test Score VA	0.076*** (0.018)	0.076*** (0.022)		-0.038 (0.033)
Female $\times$ Boy-Specific Test Score VA	0.071*** (0.017)	0.090*** (0.021)		-0.001 (0.029)
Male $\times$ Girl-Specific Test Score VA	0.000 (0.018)	0.005 (0.023)		0.016 (0.034)
Female $\times$ Girl-Specific Test Score VA	0.047*** (0.017)	0.056*** (0.022)		0.034 (0.030)
Male $\times$ Boy-Specific Course Grade VA		0.001 (0.024)	0.073** (0.030)	0.097** (0.038)
Female $\times$ Boy-Specific Course Grade VA		-0.038 (0.024)	0.005 (0.025)	0.000 (0.032)
Male $\times$ Girl-Specific Course Grade VA		-0.009 (0.026)	0.062** (0.031)	0.057 (0.040)
Female $\times$ Girl-Specific Course Grade VA		-0.019 (0.026)	0.067** (0.028)	0.040 (0.036)
Constant	-0.153 (0.182)	-0.153 (0.182)	-0.125 (0.181)	-0.125 (0.182)
Mean	0.092	0.092	0.077	0.077
N	332883	332883	332883	332883
r^2	0.769	0.769	0.465	0.465

Note: Standard errors clustered at the fifth-grade teacher level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table E.2: Gender-Differentiated Impacts of Fifth-Grade Teachers on Middle School Math and Reading Outcomes

	Test Scores		Course Grades	
	I	II	III	IV
<i>Panel A: Math Outcomes</i>				
Male × Boy-Specific Math Score VA	0.070*** (0.020)	0.060*** (0.023)		-0.016 (0.033)
Female × Boy-Specific Math Score VA	0.094*** (0.018)	0.109*** (0.022)		0.041 (0.029)
Male × Girl-Specific Math Score VA	0.021 (0.019)	0.018 (0.023)		0.018 (0.032)
Female × Girl-Specific Math Score VA	0.050*** (0.018)	0.061*** (0.022)		0.012 (0.029)
Male × Boy-Specific Math Grade VA		0.023 (0.026)	0.065** (0.032)	0.072* (0.040)
Female × Boy-Specific Math Grade VA		-0.032 (0.025)	0.010 (0.027)	-0.020 (0.033)
Male × Girl-Specific Math Grade VA		0.008 (0.028)	0.081** (0.032)	0.071* (0.039)
Female × Girl-Specific Math Grade VA		-0.026 (0.027)	0.072** (0.030)	0.054 (0.037)
Constant	-0.187 (0.191)	-0.188 (0.191)	-0.334 (0.204)	-0.333 (0.204)
Mean	0.101	0.101	0.077	0.077
N	332,883	332,883	332,883	332,883
R ²	0.707	0.707	0.389	0.389
<i>Panel B: Reading Outcomes</i>				
Male × Boy-Specific Reading Score VA	0.069*** (0.022)	0.063** (0.025)		0.005 (0.034)
Female × Boy-Specific Reading Score VA	0.051** (0.021)	0.066*** (0.024)		-0.009 (0.029)
Male × Girl-Specific Reading Score VA	0.014 (0.022)	0.014 (0.026)		0.023 (0.037)
Female × Girl-Specific Reading Score VA	0.073*** (0.021)	0.060** (0.025)		0.049 (0.031)
Male × Boy-Specific Reading Grade VA		0.015 (0.027)	0.093*** (0.032)	0.086** (0.037)
Female × Boy-Specific Reading Grade VA		-0.035 (0.027)	0.015 (0.028)	0.013 (0.032)
Male × Girl-Specific Reading Grade VA		-0.001 (0.029)	0.059* (0.033)	0.043 (0.040)
Female × Girl-Specific Reading Grade VA		0.030 (0.029)	0.076** (0.030)	0.046 (0.037)
Constant	-0.037 (0.235)	-0.039 (0.235)	0.130 (0.207)	0.132 (0.207)
Mean	0.083	0.083	0.076	0.076
N	332,883	332,883	332,883	332,883
R ²	0.675	0.675	0.387	0.387

Note: Standard errors clustered at the fifth-grade teacher level. * $p < 0.10$,

** $p < 0.05$, *** $p < 0.01$.

F Additional Results

F.1 Subject-Specific Multidimensional Impacts

Table F.1 reports the multidimensional value-added regressions estimated in Section 5.1 separately for math (Panel A) and reading (Panel B). For each subject, the table reports impacts on test scores and course grades across two outcome windows (sixth-eighth grade and seventh-eighth grade), with specifications that include test score value-added alone, course grade value-added alone, and both together. The discussion in Section 5.1 summarizes the substantive findings: own-outcome effects are similar across subjects, while patterns in cross-impacts differ across subjects.

Table F.1: Multidimensional Impacts of Fifth-Grade Teachers on Middle School Math and Reading Outcomes

<i>Panel A: Math Outcomes</i>								
	Math Scores (6th-8th)		Math Scores (7th-8th)		Math Grades (6th-8th)		Math Grades (7th-8th)	
	I	II	III	IV	V	VI	VII	VIII
Test Score VA	0.168*** (0.019)	0.160*** (0.019)	0.131*** (0.020)	0.124*** (0.021)		0.078*** (0.025)		0.016 (0.028)
Course Grade VA		0.051* (0.031)		0.049 (0.034)	0.193*** (0.042)	0.162*** (0.044)	0.144*** (0.046)	0.138*** (0.048)
Constant	-0.189 (0.191)	-0.189 (0.191)	-0.075 (0.206)	-0.075 (0.206)	-0.326 (0.203)	-0.323 (0.203)	-0.379 (0.243)	-0.378 (0.243)
Mean	0.099	0.099	0.098	0.098	0.077	0.077	0.072	0.072
r ²	0.707	0.707	0.699	0.699	0.388	0.388	0.376	0.376
<i>Panel B: Reading Outcomes</i>								
	Reading Scores (6th-8th)		Reading Scores (7th-8th)		Reading Grades (6th-8th)		Reading Grades (7th-8th)	
	I	II	III	IV	V	VI	VII	VIII
Test Score VA	0.182*** (0.034)	0.185*** (0.034)	0.137*** (0.038)	0.138*** (0.038)		0.085* (0.044)		0.064 (0.050)
Course Grade VA		-0.016 (0.033)		-0.007 (0.037)	0.119*** (0.044)	0.108** (0.045)	0.138*** (0.050)	0.129** (0.051)
Constant	0.000 (0.235)	0.000 (0.235)	0.053 (0.266)	0.053 (0.266)	0.170 (0.209)	0.171 (0.209)	0.375 (0.250)	0.375 (0.250)
Mean	0.082	0.082	0.080	0.080	0.075	0.075	0.070	0.070
N	334425	334425	222950	222950	334425	334425	222950	222950
r ²	0.675	0.675	0.663	0.663	0.386	0.386	0.380	0.380

Note: Standard errors clustered at the fifth-grade teacher level. Test Score VA is defined on fifth-grade test scores. Course Grade VA is defined on sixth-grade course grades. All outcomes are standardized (z-scores). Models include fourth-grade classroom fixed effects and fifth-grade school fixed effects. Stars denote significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

F.2 Relative Skills Framework Test: All Outcomes

Table F.2 reports the relative-skills framework tests estimated in Section 5.3 across all four 7th-8th grade outcomes: math scores, math grades, reading scores, and reading grades. The body table (Table 6) reports the two outcomes with the largest gender-differentiated effects (math scores and reading grades); the appendix table reports the full set. For each outcome, the table reports three specifications: a gender-differentiated baseline (columns I, IV), a

skill-heterogeneity test (columns II, V), and a specification combining both tests (columns III, VI). The substantive findings are summarized in [Section 5.3](#).

Table F.2: Relative Proficiency Test: Seventh-Eighth Grade Outcomes

<i>Panel A: Math Outcomes</i>						
	Math Scores			Math Grades		
	I	II	III	IV	V	VI
Female	0.218*** (0.080)	0.056*** (0.003)	0.217*** (0.080)	0.455*** (0.108)	0.302*** (0.004)	0.435*** (0.107)
TVA	0.089*** (0.029)	0.129*** (0.021)	0.091*** (0.029)	0.190*** (0.069)	0.145*** (0.046)	0.186*** (0.068)
Score RP		-0.001 (0.004)	-0.001 (0.004)		-0.118*** (0.006)	-0.117*** (0.006)
Female × TVA	0.084** (0.040)		0.074* (0.040)	-0.093 (0.094)		-0.083 (0.095)
Score RP × TVA		-0.093*** (0.035)	-0.087** (0.035)		0.093 (0.077)	0.088 (0.077)
Constant	-0.073 (0.205)	-0.071 (0.206)	-0.070 (0.204)	-0.383 (0.242)	-0.368 (0.239)	-0.374 (0.238)
Mean	0.098	0.098	0.098	0.072	0.072	0.072
N	222,950	222,950	222,950	222,950	222,950	222,950
R ²	0.699	0.699	0.699	0.377	0.378	0.379
<i>Panel B: Reading Outcomes</i>						
	Reading Scores			Reading Grades		
	I	II	III	IV	V	VI
Female	-0.511*** (0.081)	0.021*** (0.003)	-0.506*** (0.081)	0.319*** (0.108)	0.301*** (0.004)	0.306*** (0.108)
TVA	0.107* (0.056)	0.136*** (0.038)	0.112** (0.056)	0.235*** (0.075)	0.135*** (0.050)	0.225*** (0.074)
Score RP		0.049*** (0.004)	0.049*** (0.004)		-0.135*** (0.006)	-0.134*** (0.006)
Female × TVA	0.061 (0.079)		0.049 (0.079)	-0.194* (0.100)		-0.180* (0.100)
Score RP × TVA		-0.126** (0.064)	-0.125* (0.064)		0.125 (0.086)	0.116 (0.086)
Constant	0.018 (0.266)	0.043 (0.268)	0.008 (0.268)	0.336 (0.249)	0.404 (0.248)	0.365 (0.248)
Mean	0.080	0.080	0.080	0.070	0.070	0.070
N	222,950	222,950	222,950	222,950	222,950	222,950
R ²	0.664	0.664	0.664	0.381	0.383	0.384

Note: Standard errors clustered at the fifth-grade teacher level. For Panel A, TVA refers to a 5th grade teacher's math score value-added (defined on 5th grade math scores) for columns I-III, and a 5th grade teacher's math grade value-added (defined on 6th grade math grades) for columns IV-VI. For Panel B, TVA refers to a 5th grade teacher's reading score value-added (defined on 5th grade reading scores) for columns I-III, and a 5th grade teacher's reading grade value-added (defined on 6th grade reading grades) for columns IV-VI. All outcomes are standardized (z-scores). Models include fourth-grade classroom fixed effects and fifth-grade school fixed effects. Stars denote significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

F.3 Analysis of High-School Outcomes

F.3.1 Decomposing the Joint Outcome

Let S_i denote an indicator for whether student i appears in the CGR sample, and let G_i denote an indicator for whether student i graduates within the relevant window. Graduation is defined only for students in the CGR sample, so G_i is observed only when $S_i = 1$. The jointly-defined outcome used in [Section 5.4](#) is the product of the two:

$$Y_i = S_i \cdot G_i \quad (\text{F.1})$$

so that $Y_i = 1$ if student i appears in the CGR sample and graduates, and $Y_i = 0$ otherwise — including for students who never appear in the sample, for whom G_i is undefined.

I estimate three specifications of [Equation \(7\)](#), each on a different outcome. The first uses S_i as the outcome and is estimated on the full analysis sample; this is Column I of [Table 7](#). The second uses Y_i and is likewise estimated on the full sample; these are Columns II and III. The third uses G_i and is estimated only on students with $S_i = 1$; these are the conditional estimates reported in [Table F.3](#) below. Denote the corresponding coefficients on teacher value-added δ_j^{VA} by $\hat{\beta}^S$, $\hat{\beta}^Y$, and $\hat{\beta}^G$ respectively.

The joint outcome's expectation factors into the two margins by the law of iterated expectations. Conditioning on teacher value-added and the covariates in [Equation \(7\)](#),

$$\mathbb{E}[Y \mid \delta^{VA}] = \Pr(S = 1 \mid \delta^{VA}) \cdot \mathbb{E}[G \mid S = 1, \delta^{VA}] \quad (\text{F.2})$$

since $Y = 1$ requires both that a student appear in the CGR sample and that they graduate once there. Differentiating with respect to δ^{VA} gives

$$\underbrace{\frac{\partial \mathbb{E}[Y]}{\partial \delta^{VA}}}_{\text{joint}} = \underbrace{\mathbb{E}[G \mid S = 1] \cdot \frac{\partial \Pr(S = 1)}{\partial \delta^{VA}}}_{\text{sample channel}} + \underbrace{\Pr(S = 1) \cdot \frac{\partial \mathbb{E}[G \mid S = 1]}{\partial \delta^{VA}}}_{\text{graduation channel}} \quad (\text{F.3})$$

As a result, the impact of a teacher on the joint outcome can be decomposed into two channels. The first term reflects the change associated with students entering the CGR sample, while the second term reflects the change in the observed graduation rate among students who enter the sample. Since teacher value-added may affect sample entry, this second term represents a descriptive accounting channel that combines both actual changes in graduation likelihood and changes in the composition of the observed students. The empirical counterparts of these derivatives

are the coefficients defined above – $\hat{\beta}^Y$, $\hat{\beta}^S$, and $\hat{\beta}^G$ – and the two scaling terms are the sample means of G among students with $S = 1$ and of S in the full sample. The decomposition therefore implies

$$\hat{\beta}^Y \approx \overline{G}_{S=1} \cdot \hat{\beta}^S + \bar{S} \cdot \hat{\beta}^G \quad (\text{F.4})$$

where $\overline{G}_{S=1}$ is the mean of G among students in the CGR sample and $\bar{S}$ is the mean of S in the full sample.²⁸

Table F.3 reports the estimates from estimating Equation (7) for graduation in four years and six years, conditional on being in the CGR sample.

Table F.3: Impacts of Fifth-Grade Teachers on High School Graduation, Conditional on CGR Sample Membership

	Graduated	
	Within 4 Years I	Within 6 Years II
Female	0.070 (0.051)	0.078* (0.042)
Test Score VA	0.049* (0.029)	0.036 (0.025)
Female × Test Score VA	-0.033 (0.039)	-0.002 (0.032)
Course Grade VA	0.028 (0.039)	0.055 (0.033)
Female × Course Grade VA	-0.003 (0.052)	-0.038 (0.042)
Constant	1.221*** (0.164)	1.115*** (0.154)
Mean	0.902	0.932
N	101141	101141
r ²	0.166	0.169

Note: Estimates of Equation (7) on graduation, restricted to students who appear in North Carolina’s cohort graduation rate (CGR) sample. These are the $\hat{\beta}^G$ estimates referenced in Equation (F.4). Standard errors clustered at the fifth-grade teacher level. Test Score VA is defined on fifth-grade test scores. Course Grade VA is defined on sixth-grade course grades. Models include fourth-grade classroom fixed effects and fifth-grade school fixed effects. Stars denote significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

To decompose the estimated joint effects, I compute the two scaling terms in Equation (F.4) separately for boys and girls, since the decomposition applies to each group’s coefficients and the underlying rates differ by gender. Table F.4 reports these means. Boys are slightly less likely than girls to appear in the CGR sample, and conditional on appearing, are less likely to graduate within either window.

²⁸The factorization in Equation (F.2) is exact for the underlying conditional expectations. Equation (F.4) is an approximation: the coefficients are estimated from three separate linear probability models on different samples, and the scaling terms are evaluated at sample means rather than at any particular value of δ^{VA} .

Table F.4: Sample Means Used in the Decomposition

	All	Boys	Girls	Difference
$\bar{S}$: In CGR sample	0.909 (0.288)	0.905 (0.294)	0.913 (0.282)	-0.008*** (0.002)
$\bar{G}_{S=1}$: Graduated within 4 years	0.902 (0.297)	0.889 (0.314)	0.915 (0.279)	-0.026*** (0.002)
$\bar{G}_{S=1}$: Graduated within 6 years	0.932 (0.251)	0.919 (0.272)	0.944 (0.230)	-0.025*** (0.002)

Note: $\bar{S}$ is computed over the full analysis sample ($N = 111,475$; 54,238 boys and 57,237 girls). $\bar{G}_{S=1}$ is computed over students who appear in the CGR sample ($N = 101,314$). Standard deviations in parentheses below each mean; standard errors of the difference in parentheses in the final column. Differences are computed as the boys' mean minus the girls' mean. Stars denote significance levels for a t-test for differences in means: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table F.5 applies the decomposition outlined in Equation (F.4) to the estimated impacts of course grade value-added, using the coefficients reported above and the means in Table F.4. I report the decomposition of the course grade value-added estimates, but one can recover the same decomposition for the test score value-added estimates using the same procedure.

Table F.5: Decomposition of Joint Graduation Effects, Course Grade Value-Added

		Sample channel	Graduation channel	Sum	Estimated $\hat{\beta}^Y$
Within 4 years	Boys	0.042	0.025	0.067	0.066
	Girls	0.006	0.023	0.028	0.030
Within 6 years	Boys	0.043	0.050	0.093	0.092
	Girls	0.006	0.016	0.021	0.022

Note: The sample channel is $\bar{G}_{S=1} \cdot \hat{\beta}^S$ and the graduation channel is $\bar{S} \cdot \hat{\beta}^G$, following Equation (F.4). Means are taken from Table F.4; $\hat{\beta}^S$ from Column I of Table 7; $\hat{\beta}^G$ from Table F.3. Estimates for girls are the sum of the main effect and the interaction with student gender. The final column reports the corresponding coefficient on the jointly-defined outcome from Columns II and III of Table 7.

For boys, the two channels contribute roughly equally to the six-year estimate: the sample channel accounts for 46% of the predicted total effect of 9.3 percentage points, and the graduation channel accounts for ~ 54%. At four years the sample channel dominates, contributing 63% of the predicted total of 6.7 percentage points. In both cases the predicted totals come within 0.2 percentage points of the estimated joint coefficients, indicating that the three sets of estimates are mutually consistent. For girls, the decomposition applies but has little to apportion: both channels are small, and their sum is 2.1 percentage points at six years, against an estimated joint effect of 2.2 percentage points.

F.3.2 Effects on High School GPA

Table F.6 reports the full set of coefficients underlying Figure 5 at each GPA threshold. Their interpretation is discussed in Section 5.4. The statistically significant coefficient on course grade value-added at the $\text{GPA} \geq 4$ threshold is not robust to neighboring cutoffs, at which the estimates are smaller and insignificant.

Table F.6: Gender-Differentiated Impacts of Fifth-Grade Teachers Across the High School GPA Distribution

	Has GPA											
	0.5	1	1.5	2	2.5	3	3.5	4	4.5	5	5.5	
Female	-0.027 (0.062)	-0.026 (0.062)	-0.027 (0.065)	-0.038 (0.070)	-0.019 (0.073)	-0.024 (0.074)	-0.046 (0.071)	-0.120** (0.061)	-0.167*** (0.044)	-0.056*** (0.021)	-0.002 (0.003)	
Test Score VA	0.026 (0.033)	0.027 (0.033)	0.029 (0.035)	0.036 (0.038)	0.079*** (0.040)	0.092*** (0.040)	0.081*** (0.038)	0.059* (0.032)	0.039* (0.023)	0.011 (0.010)	-0.000 (0.001)	
Female × Test Score VA	-0.022 (0.047)	-0.023 (0.047)	-0.014 (0.049)	0.007 (0.052)	-0.011 (0.055)	0.004 (0.055)	0.035 (0.054)	0.013 (0.047)	0.029 (0.034)	-0.010 (0.015)	-0.001 (0.001)	
Course Grade VA	0.131*** (0.046)	0.131*** (0.046)	0.068 (0.048)	0.015 (0.051)	-0.001 (0.055)	0.079 (0.055)	0.050 (0.052)	0.115*** (0.044)	0.051 (0.032)	0.022 (0.015)	-0.000 (0.001)	
Female × Course Grade VA	-0.128** (0.062)	-0.128** (0.062)	-0.136** (0.065)	-0.077 (0.068)	-0.030 (0.075)	-0.094 (0.077)	-0.010 (0.074)	-0.095 (0.066)	-0.045 (0.046)	-0.016 (0.021)	-0.000 (0.001)	
Constant	0.993*** (0.165)	0.993*** (0.165)	1.014*** (0.164)	1.026*** (0.160)	0.955*** (0.167)	0.875*** (0.170)	0.820*** (0.119)	0.565*** (0.108)	0.200*** (0.043)	0.023 (0.015)	0.001* (0.001)	
Mean	0.761	0.759	0.739	0.684	0.589	0.471	0.341	0.211	0.093	0.016	0.000	
N	111317	111317	111317	111317	111317	111317	111317	111317	111317	111317	111317	
r ²	0.328	0.328	0.321	0.313	0.311	0.311	0.303	0.271	0.208	0.139	0.078	

Note: Each column reports estimates of Equation (7) on a jointly-defined indicator. For the threshold columns, the indicator takes the value one if a student appears in the CGR sample *and* has a recorded weighted high school GPA at or above the stated cutoff, and zero otherwise. The “Has GPA” column uses an indicator for appearing in the CGR sample and having any recorded GPA. These estimates are illustrated in Figure 5, which plots impacts for boys and girls separately; the estimates for girls in the figure are the sum of the main effect and the interaction with student gender reported here. The ≥ 5 and ≥ 5.5 columns are omitted from the figure, as almost no students have a weighted GPA above these cutoffs (means 0.016 and 0.000). Test Score VA is defined on fifth-grade test scores. Course Grade VA is defined on sixth-grade course grades. Standard errors clustered at the fifth-grade teacher level. Models include fourth-grade classroom fixed effects and fifth-grade school fixed effects. Stars denote significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.